



**A REPORT SUBMITTED TO THE COLLEGE OF HEALTH
SCIENCES, UNIVERSITY OF ILORIN, IN PARTIAL FULFILLMENT
OF THE MINIMUM LEARNING OBJECTIVES OF THE 500 LEVEL
COBES POSTING**



THE WORLD HEALTH ORGANIZATION THEME FOR THE YEAR

2025:

“HEALTHY BEGINNINGS, HOPEFUL FUTURE”

WRITTEN BY: SHAO MEDICAL LABORATORY SCIENCES COBES

GROUP 2024/2025

MARCH, 2025.

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LABORATORY SCIENCE COBES POSTING**

BY

**MLS SET 2024/2025 SET
ACADEMIC SESSION 2024/2025**

MARCH, 2025.

ATTESTATION

This is to attest that this report was conducted and written by Medical Laboratory Science students 24/25 Set, Department of Medical Laboratory Science, College of Health Sciences, University of Ilorin, Ilorin, Kwara State, Nigeria, under our supervision. This report is adequate in both scope and content for the fulfilment of the Minimum Learning Objectives (MLOs).

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SUMMARY

The Community-Based Experience and Services (COBES) program, initiated by the College of Health Sciences at the University of Ilorin, aims to immerse medical laboratory science students in real-world community health settings. This report presents the activities and findings from a COBES posting at Shao community, Kwara State, from March 17 to March 28, 2025, involving 68 final-year medical laboratory science students. The primary objective was to evaluate local health practices, identify gaps in healthcare services, and propose sustainable initiatives to address community health issues. Through structured questionnaires and direct engagement with community members, the students gathered valuable data despite encountering challenges such as language barriers and limited access to technology. The findings reveal a community with limited healthcare options, relying heavily on informal practitioners. Recommendations include enhancing healthcare education, improving service delivery, and fostering collaborations with local stakeholders. This experiential learning opportunity significantly contributed to students' understanding of public health dynamics and reinforced the importance of community-based interventions in medical education. The outcomes underscore the need for continuous improvement of health services and the promotion of health awareness within the community.

INTRODUCTION TO COBES

The Community Based Experiences and Services (COBES) programme, initiated by the College of Health Sciences, University of Ilorin, is an encompassing, multi-disciplinary initiative. It is designed to immerse medical laboratory students within community settings, thereby creating a deep understanding of local health practices, inadequacies and needs with the purpose of drawing conclusions and implementing initiatives to solve observed needs within such communities.

This program commenced in the then Faculty of Health Sciences of the University in 1978, stemming from the vision of the Faculty to pursue a different academic curriculum which will be *Student-centered, Community-based and Problem-solving*. It focuses on the ability of the students to enter into a new community, interact with people, work with each other, identify problems and provide acceptable and working solutions.

Our class was assigned a single location within the state of Kwara. Our group, comprising of 68 students, was sent to a community on the outskirts of Ilorin known as Shao.

Every year, the COBES posting is always fashioned toward the World Health Day theme for the year. The World Health Day is celebrated on 7 April to mark the anniversary of the founding of the World Health Organization (WHO) in 1948. World Health Day is a global campaign, inviting everyone—from global leaders to the public in all countries to focus on a single health challenge with global impact.

The World Health Day theme for 2025 year is "**Healthy beginnings, hopeful futures.**" It aims to Reduce preventable deaths, Improve care quality, Support women's long-term health, Encourage collective action to mobilize communities and health systems to share vital health information and work together to build healthier families and communities. 7 Minimum Learning Objectives (MLOs) were extracted from the theme. The Minimum Learning Objectives are:

1. To extract data on the number and type of health facilities and services present in within the community.
2. To determine the various challenges mitigating easy access to satisfactory health services.
3. To determine the impediment inherent with the health services providers and facilities making access to satisfactory health care, a daunting task.
4. To assess the level of government and private sector involvement in health provision in the community.
5. To sensitize the people and community on the importance and proactive measures to facilitate health promotion.
6. To mobilize community resources in provision of at least one project that will enhance.
"It is time to build a fairer. Healthier world for Everyone, Everywhere"
7. Write a report and present same to the provost, College of Health Sciences, University of Ilorin.

COMMUNITY ENTRY AND INITIAL ACTIVITIES

1.1 Departure from College of Health Sciences

Our COBES posting commenced on Monday, 17th of March, 2025. We departed from the College of Health Sciences (New College) to Shao community at 9:30am using transportation provided by the university. We were conveyed to our destination on 4 different buses, this led to different arrival time. The journey lasted for about an hour with the first 3 the first buses arriving at about 10:30am and the last buses arriving few minutes later at about 10:45am.

On our way to our hall of residence, we took cognizance of the Township market, the King's palace, Shao community school, the Central Mosque and other notable places in the community. Few minutes after our arrival at the COBES site, our focal person, Elder Job Bello (typically called Baba Bello) as well as the security officer for the COBES site came around to receive and welcome us into the community.

We had some minutes of interaction where we were introduced to Shao community, and given important information about the community including instructions for our effectiveness and safety. We made known to him our intention to see and inform the king of our arrival into the community, and other important wards within the community.



Figure 1.1: Departure from university grounds and entry into Shao community.

1.2 Visitation to the King's Palace

After cleaning our residence, three delegates headed by the group leader and accompanied by Baba Bello went to see the king, His Royal Highness, the Ohoro of Shao, Oba Dr. Job Obalowu Atolagbe Ilufemiloye Ayantola IV for the first time at 4:00pm. We introduced ourselves and informed him of the reason for our presence and our plans for the community. We informed him of our intention for the entire group to come and pay him a visit that same day, to which he instructed us to come the next day between 11am to 1pm. We obliged his directives and took our leave.



Figure 1.2: Initial Visit to the King

The next day, Tuesday, 18th of march, 2025, the entire group set out to pay the king a visit at exactly 11am, Unfortunately, the king informed us that one of the chiefs in his cabinate passed away overnight and he would unable to attend to us. He instructed that the whole group should come the following day. We paid our condolenses and took our leave.

1.3 Official Visit to the King

On Wednesday, 19th of February 2024, the entire COBES group had its official visit to the king. By 11:45am, we arrived at the king's palace.



Figure 1.3: Official visit to the king

At the King's palace, we were welcomed warmly by him. He admonished us to work very hard, remain focused and make adequate plans to serve the people. He appealed to us to help build a better version of Nigeria's Healthcare. Importantly, he informed us of the security details of the community. He also told us of certain rules and traditions of the community which we should adhere to.

Within the palace, the historical perspective of Shao was requested by one of us. At the king then instructed the Elemosho Amona of Shao, Chief Joseph Oyerinde to narrate the history of Shao. The Elemosho narrated in english, enlightening overview of the town's cultural identity, encompassing its rich history, prominent landmarks, and cherished festivals, thereby broadening our knowledge and appreciation of the local community.

The king also spoke about some of the rites of coronation done by every newly appointed kings, and named every king to sit on the throne of Shao town. We informed him of our intentions to carry out awareness outreaches in various schools within the community and a medical outreach at the Primary Healthcare Centre. He was glad and approved of the plans stating that for whatever we plan to do, we can count on the people of the community for maximum participation

1.4 Official Visit to the Primary Health Care Centre, Shao

On Tuesday, 18th of march 2024, at 9:00am, we arrived at the Primary Healthcare Centre for the first time. We introduced ourselves and were received by the Head Nurse S.B Ibrahim who welcomed us heartily and helped in showing us around the PHC. We were then shown the laboratory and the scientist present in person of scientist Saduu Ibrahim Kolawale. We spent about 2hours speaking with the scientist who seemed very interested in the MLS program at the university of ilorin. He asked many questions and we provided as much answers as we could. We also had the opportunity to ask some relevant questions needed for our survey.



Figure 1.4: Official visit to the Primary Health Center.

1.5 Visit to the Wards in the Community

During the early days of our posting, we were taken on a tour of the community by Baba Bello. We were shown all the 8 wards of Shao, identifying the boundaries and notable features of each ward. We were also shown several township monuments and some of which includes the following:

- The house of the great warrior of Shao town (the Ajakitipa)
- The town deities including Odu, Banja and Awon.
- The 4 mountains surrounding the community, which we decided to hike one of the mountains.
- The Awon river and so on.

We also went to different schools and religious establishments to inform them of our outreach plans and the proposed date of our coming.



Figure 1.5: Tour of Shao Community.

SHAO TOWNSHIP: HISTORY AND SOCIO-DEMOGRAPHY

2.1 History of Shao

Shao Awonga's history is a vibrant tapestry woven from the threads of tradition, resilience, and unity. Located approximately 12 kilometers away from Ilorin, Shao resides within the Moro Local Government Area of Kwara state.

According to local lore, in the 17th century, four hunters settled independently at the base of hills in various locations. These pioneering settlers were Ajanki, Iboo, Papala, and Ajakitipa. They ventured from Old Oyo empire. These four hunters settled independently at the base of hills in various locations. Their paths crossed during a hunting expedition, leading them to merge into a single community for mutual survival. Thus, the name '*Saho*' emerged from the amalgamation, combining '*sa*' meaning '*to gather*' and '*ho*' meaning '*to make noise or to chat together*'. Over time, the name evolved into **Shao**.

A turning point in the early days of Shao Awonga came with the mysterious visit of a one-breasted woman. Arriving via Iboo, she spent several days among the settlers and, before her departure, delivered a profound message: the community should annually commemorate her visit by uniting in a mass wedding, blessing marriages among their children on the same day. In an act that deepened the local mythos, water gushed from the spot where she vanished, giving rise to the tradition of drawing water with one's hands—known as *Awanmu*—and inspiring the naming of the Awon mass wedding. This celebration, typically held between late October and early December (with the most recent on November 3rd), has become a hallmark of communal unity and prosperity.

In its quest to reconnect with its heritage, the people of Shao Awonga sought the blessing of the then Alaafin of Oyo, Alaafin Abiodun Olabisi. Responding to this call, after three visits, a prince known as Olanibo Olosha Oko was sent and appointed as a wise man, Ohoro of Shao, to settle disputes and guide the community. The legacy continues with the

current kabiyesi, the 11th Ohoro of Shao, Oba Dr. Job Obalowu Atolagbe, who ascended the throne on 24th October, 2023. His reign is marked by a strict code of conduct: the king must not dine outside the palace, must keep his head covered at all times, is forbidden from removing his shoes outside the palace, and is required to visit specific parts of Shao after the sixth day of his rites, during which pigeons are used symbolically to wash his head and legs.

Shao Awonga is also distinguished by its rich cultural festivals and traditions. The community celebrates a variety of events including the biennial Super Egungun festival (Ikosin), the annual Egungun before the rainy season, the Sango festival in July or August, as well as the Ifa, Orisaoko, and Ogun festivals—the latter of which sees offerings of dogs, beer, and palm wine. These festivals not only celebrate the past but also reinforce the community's bonds and shared identity.

Over the years, Shao Awonga has evolved into a community marked by religious and occupational diversity. Its residents practice Traditional religion, Islam, and Christianity in a spirit of peaceful coexistence. While the forefathers were primarily hunters and farmers, today the community boasts a wide range of professions including agriculture, civil service, and various other modern pursuits. This dynamic evolution mirrors the community's steadfast connection to its roots in Old Oyo, which continues to influence its governance and cultural practices.

Geographically, Shao Awonga is nestled among majestic hills and traversed by numerous rivers. Notable waterways include the River Moro from Ilorin, River Awon to the north, and River Busamu towards Malete. The landscape is further defined by hills such as Mogba, Yelu, Oke Ikoko, Oke Isa, Oke Adaba, and Sobi Hill. Traditionally divided into eight wards or sections—Oju Oja, Isale Oja, Ode Are, Ode Alawon, Ode Lade, Ogidiri, Oke Boo, and Oke Sinniga—the community has since expanded into 41 divisions, each

contributing to the distinct identity of Shao. The indigenous Yoruba spoken here remains undiluted, and older community members proudly display tribal marks such as Abaja, Keke, Pele and Bamu.

An integral part of Shao Awonga’s security and cultural fabric is the revered role of protectors. **Ajakitipa**, known as the supreme warrior, stands as the guardian who ensures that the citizens are shielded from slavery and external threats. In addition, three mystical protectors are believed to watch over the community. The first, **Odu**, is traditionally seen in front of the king’s palace. The second protector, **Awon**—a female deity with one breast—resides at Ode Alawon, while the third, **Banja**, is associated with Oke Sinniga. These entities are believed to manifest in human form at specific times to guide and protect Shao, their presence echoing in the *oriki* (praise poetry) of the community. Odu has been reported to be seen physically in human form in 1958 and 1968 by a resident of Shao.

In essence, the story of Shao Awonga is one of convergence—of histories, cultures, and traditions. From its tumultuous beginnings to its flourishing present, the community’s narrative is a celebration of unity, identity, and the enduring power of collective heritage.

2.2 Chieftancy in Shao Town

TITLE	PRESENT TITLE HOLDERS
High Chief Ajankii Of Shao	Chief Joshua Alawode Oyedokun
High Chief Ajakitipa Of Shao	Chief Saka Oyediran
High Chief Papala Of Shao	Chief Jimoh Bolakale Rufai
High Chief Iboo Of Shao	Chief Bolaji Oyelowo Ominrinde
Elemosho Okomi Of Shao	Chief Henry Afolabi Popoola
Elemosho Amona Of Shao	Chief Joseph Oyerinde
Iya-Ilu Of Shao	Chief Mrs. Ruth Adeyola Olatunji
Otun-Ilu Of Shao	Chief Mrs. Oluwatoyin James Ramonu

2.3 The Cognomen of Shao (Oriki Ilu Shao)

ORIKI ILU SHAO	COGNOMEN OF SHAO TRANSLATED TO ENGLISH
Shao Awonga	Shao Awonga
Omo Onibu eja	Children of deep river fish
Isin sebo sebo	Fingerlings sacrifice, sacrifice
Bii kodo ma gbe	That river does not dry
Ikoru sebo	Common carp fish sacrifices
Bii kodo maa-fa	That river does not recede
Atisin ati ikoro	Both fingerlings and common carp fish
Lao fi jo orunla	We will use to prepare dried okro soup
Nile Baba to bi wa lomo	In the house of our father that born us
Awa lomo Sobi	We are the children of Sobi
Olohun meje	With seven voices
Sinniga abohun mefa	Sinniga with six voices
Idi opoto	The trunk of baobab tree
Abohun merindilogun	With sixteen voices

2.4 Sociodemographic Survey

The Shao community displays socio-demographic features that reflect its cultural background, economic conditions, and educational achievements. The community was organized into 8 teams, with each team comprising 8-9 members. Each team ventured into various wards within the community to conduct surveys using Google Forms. A team member filled out the questionnaire based strictly on the responses provided by the responder. A total of 497 individuals participated in the survey, which was utilized to extract pertinent socio-demographic and health data.

Based on the collected information—encompassing factors such as age, gender, occupation, religion, ethnicity, household income, marital status, and education level—this section offers a comprehensive overview of the community.

2.4.1 Age Distribution

Out of 497 respondents, the age structure of the Shao community skews toward younger adults, with 61% of the population falling below 40 years of age. This indicates a potentially vibrant workforce and highlights the importance of youth-oriented policies, particularly around employment and skills acquisition. 39% of the population fall within older age brackets (40 years and above), suggesting a lower dependency ratio but also emphasizing the need for health care and social support for seniors.

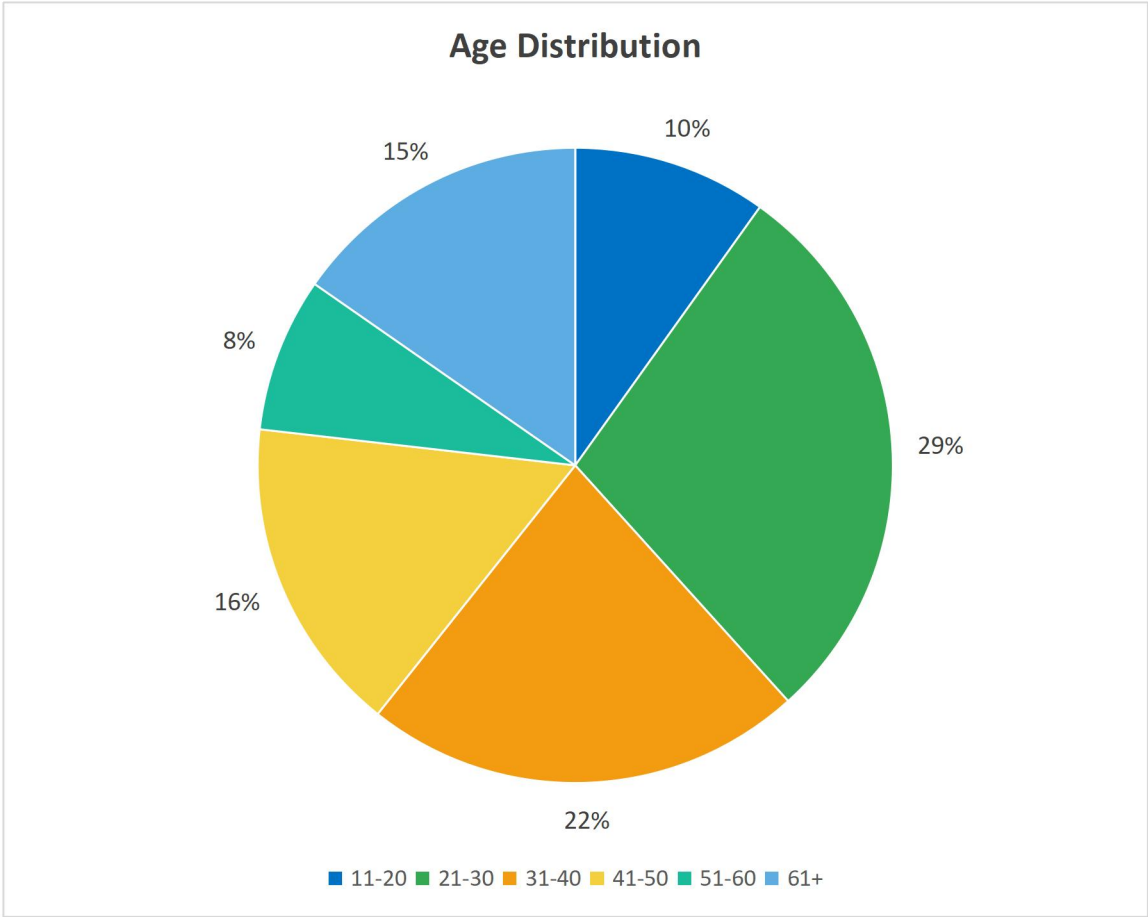


Figure 2.1: Age distribution in Shao

2.4.2 Sex Distribution

In terms of sex distribution, the data reflect a relatively balanced but slightly female-majority population. This slight skew has implications for local labor markets, household structures, and community leadership roles.

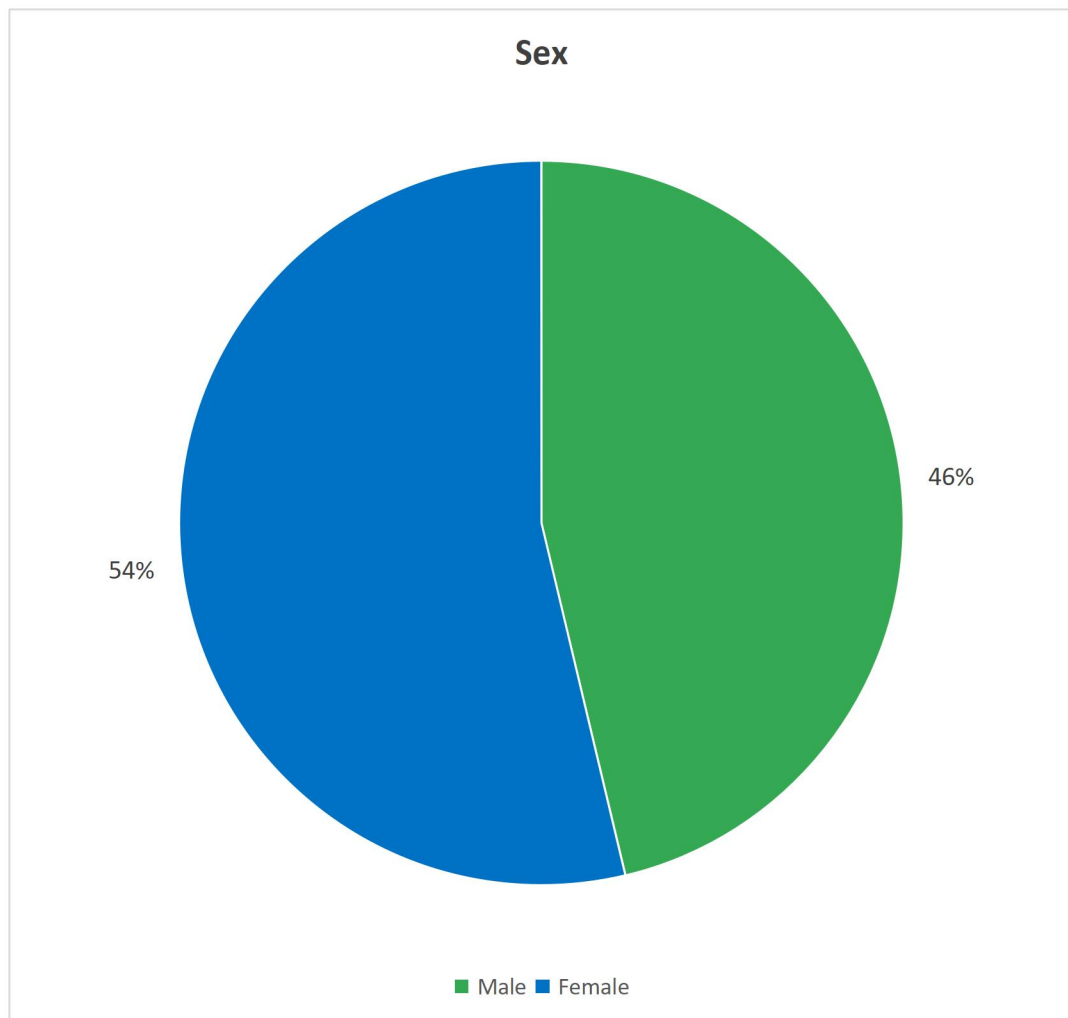


Figure 2.2: Sex Disitribution in Shao

2.4.3 Occupation and Income Levels

Occupational data from Shao reveal a diversified economy. Farming and trading appear to be primary economic activities, which is typical of many semi-urban or rural communities in the region. Civil service positions and artisanal/skilled work also make up a notable share, suggesting that a portion of residents benefit from formal employment or vocational training. “Other” categories—consisting of hunters, road transport workers, community health workers, and other informal services—reflect the prevalence of small-scale entrepreneurship.

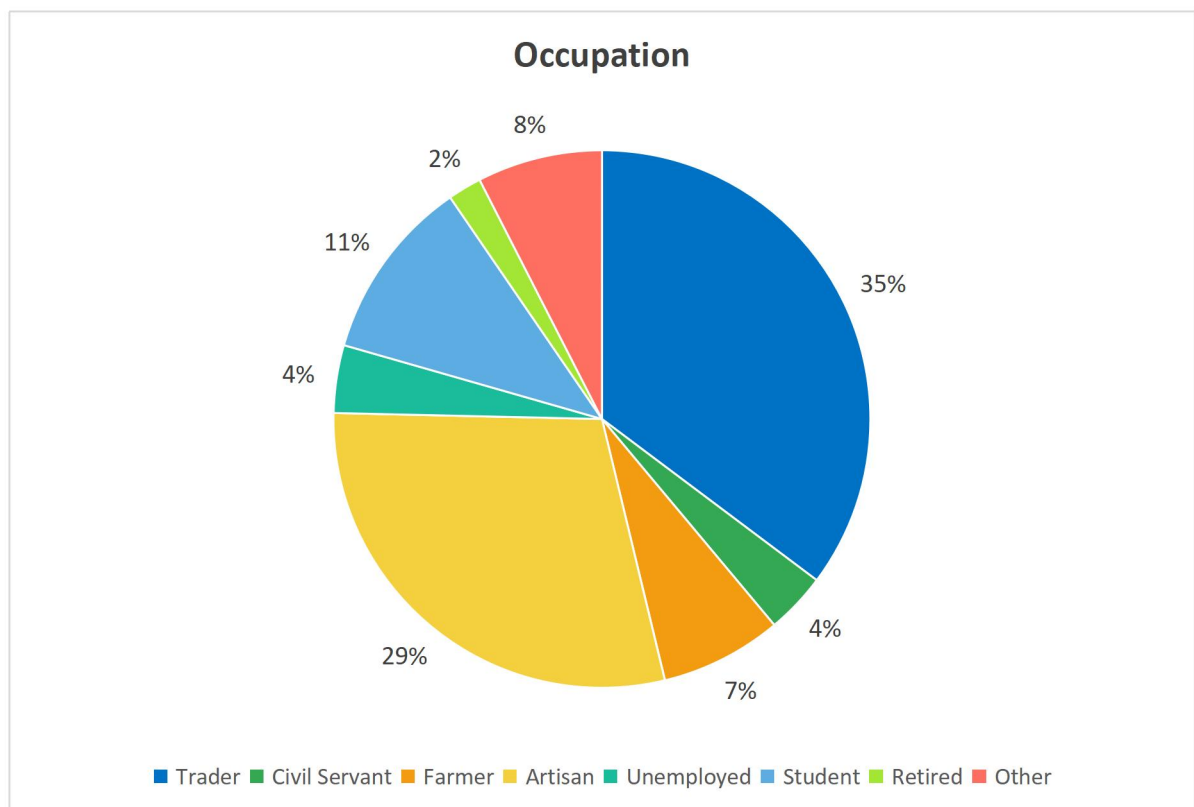


Figure 2.3: Occupation of members of the community

Household income levels tend to cluster in the low-to-mid ranges, although there is a modest group earning higher incomes. A significant segment (87%) of the community earns within low income brackets (e.g., between ₦0 and ₦50,000 per month) and moderate income brackets (e.g., between ₦50,000 and ₦100,000 per month). The presence of households with incomes above ₦100,000 suggests that a minority enjoy relatively higher earning power—often linked to stable employment or well-established enterprises.

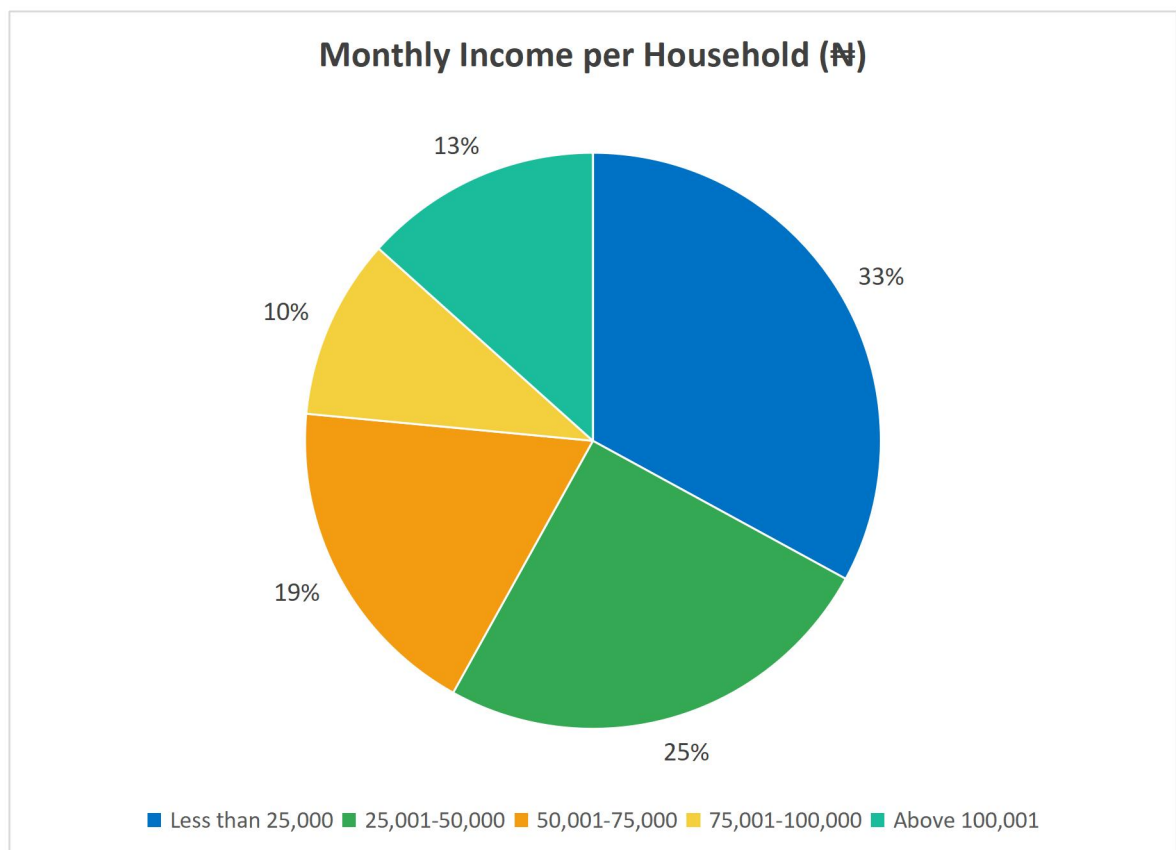


Figure 2.4: Income level of members of the community.

2.4.4 Ethnicity, Religion, and Marital Status

The Shao community is largely composed of one dominant ethnic group—often Yoruba in many parts of Kwara State—alongside smaller groups such as Hausa, Igala, Tiv, Edo, Basa, and others.

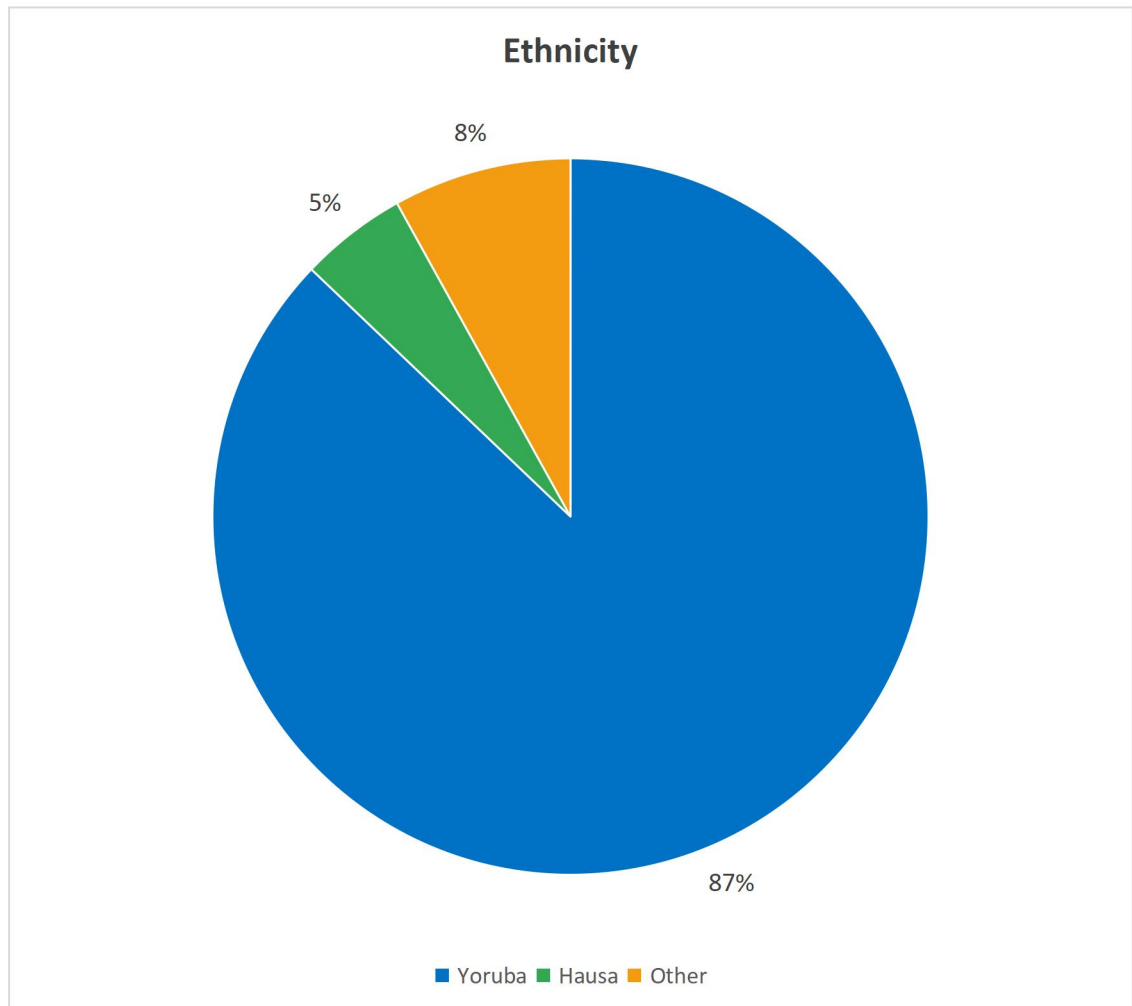


Figure 2.5: Ethnicity distribution

Religious affiliation shows a majority following Christianity or Islam, with a smaller percentage adhering to Traditional religion. This religious diversity points to a community that values pluralistic beliefs. Interfaith dialogue and collaboration is highly encouraged to maintain harmony and leverage the collective strengths of different religious groups for community development

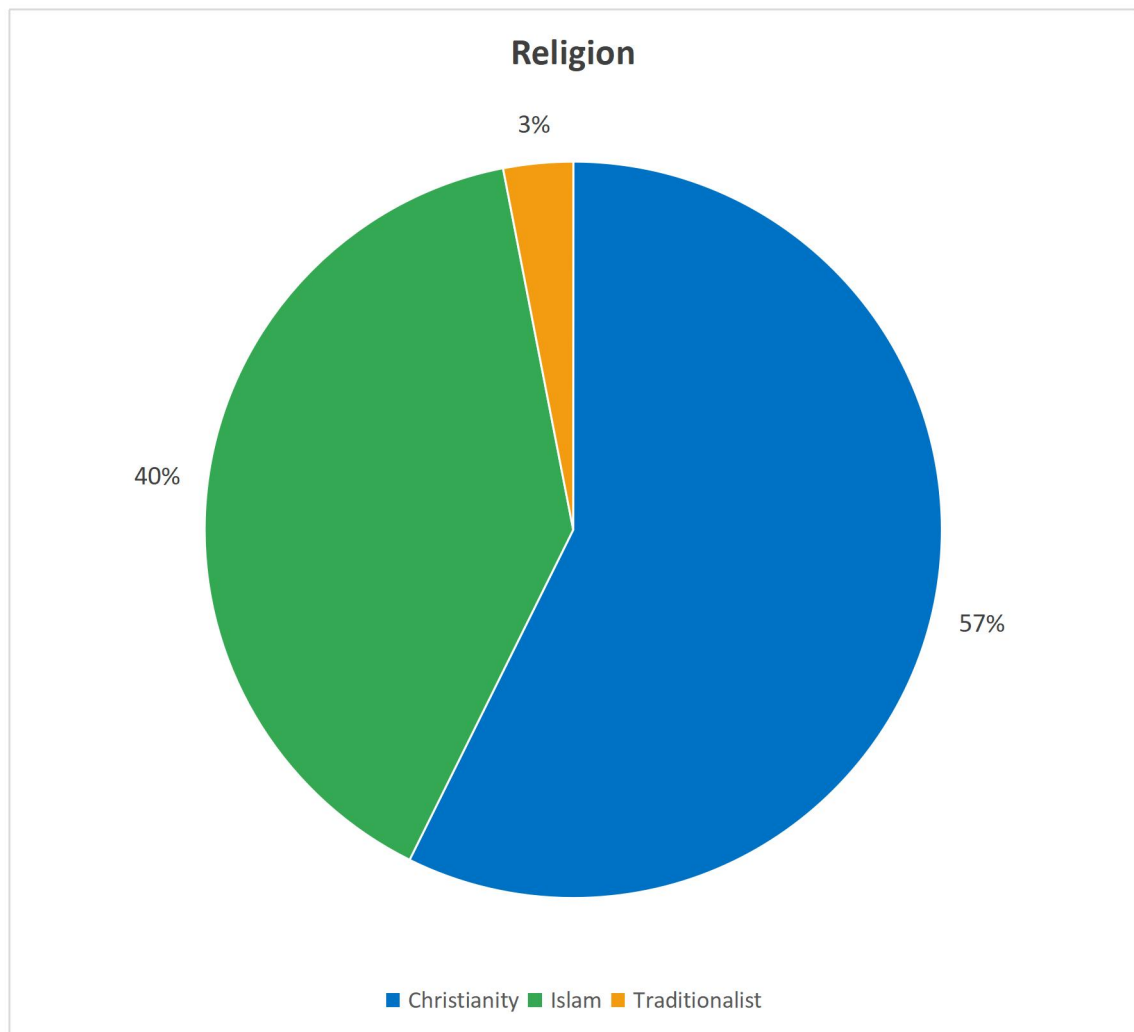


Figure 2.6: Religion distribuion

Marital status data reveal that a significant portion of the adult population is married, reflecting the traditional social structure in which marriage and family units are highly valued.

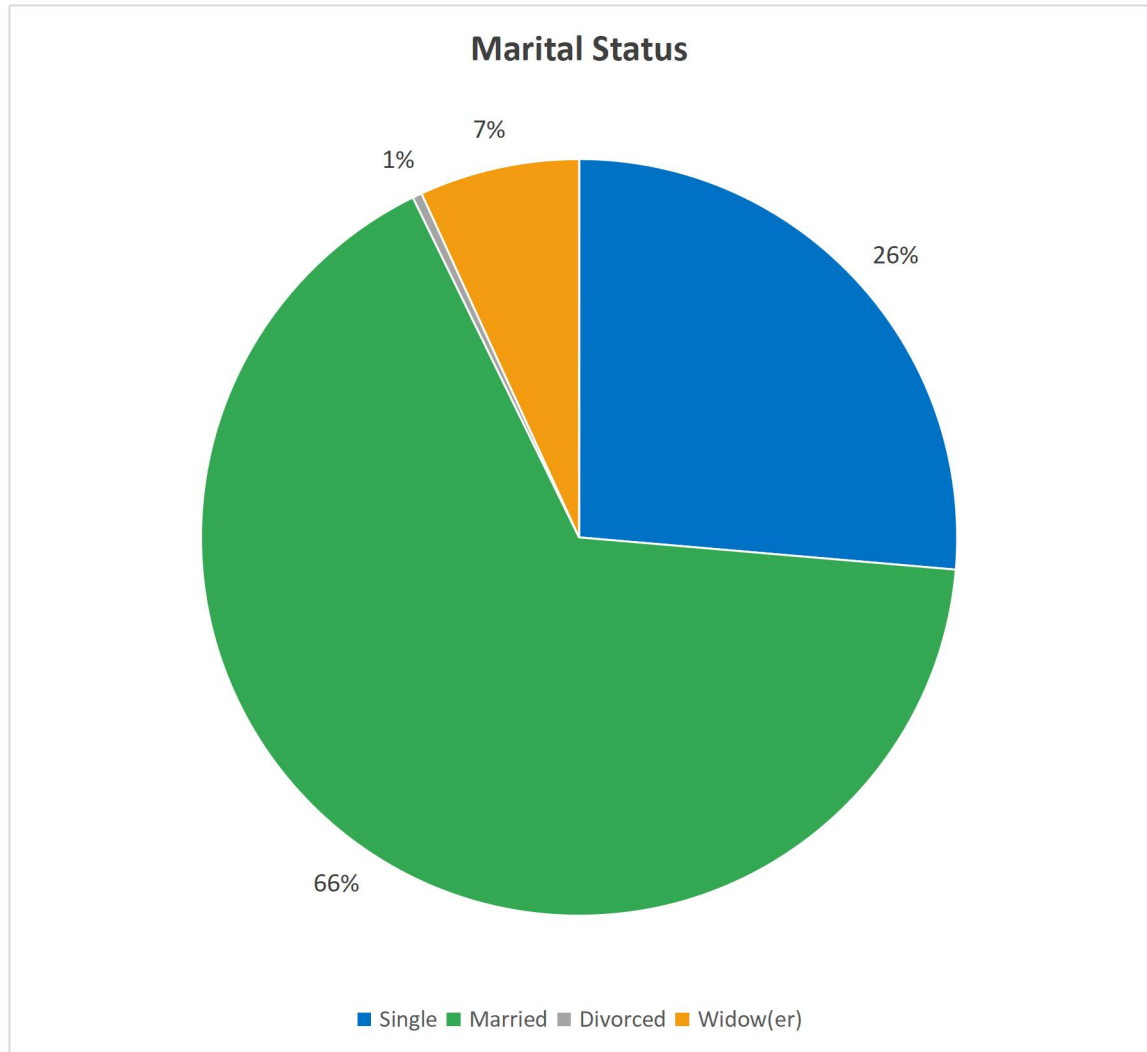


Figure 2.7: Marital status of the members of the community.

2.5 Education

The level of educational attainment within Shao spans from primary and secondary education to tertiary qualifications and vocational training. A considerable share (about 63%) of the population has completed at least a secondary-level education, which bodes well for basic literacy and employability. However, the portion of residents with tertiary education or specialized vocational skills remains smaller (27%). Despite these data aforementioned, a notable amount of the population are either completely illiterate (21%), or primary school certificate holders (16%).

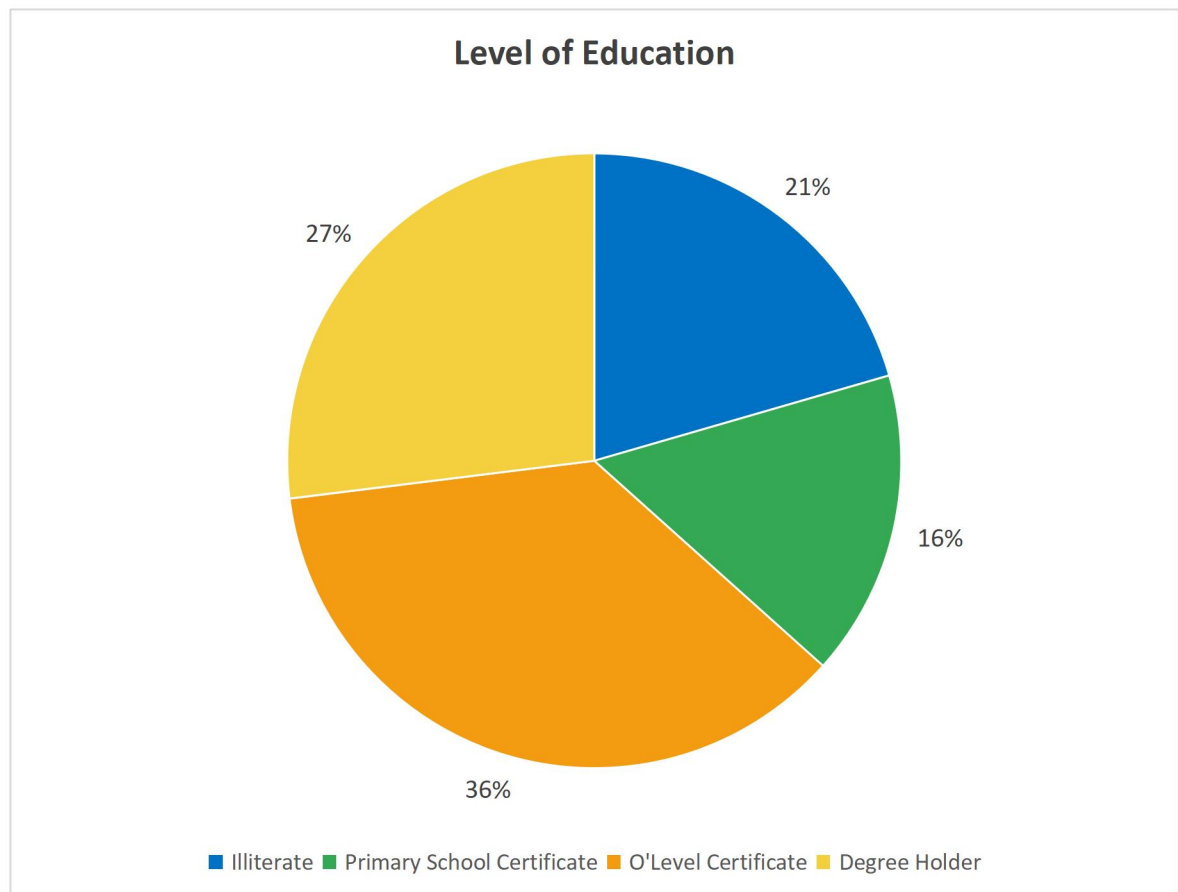


Figure 2.8: Literacy levels of members of the community.

2.6 Waste Disposal Systems

The data collected from 497 respondents reveal a sanitation system in Shao that is heavily skewed towards open defecation, with 44% of residents practicing it. This high percentage raises significant public health concerns, including the spread of waterborne diseases and environmental contamination. In contrast, 37.1% of the population uses water closets, indicating that while a considerable segment of the community has access to modern sanitation facilities, nearly half of the residents remain at risk due to inadequate waste disposal practices. Meanwhile, only 18.9% of residents use pit latrines, suggesting that this method is not widely adopted. The overall findings underscore a critical need for improved sanitation infrastructure, enhanced community education, and targeted policy interventions to reduce the prevalence of open defecation and to promote healthier, more sustainable waste disposal methods in Shao.

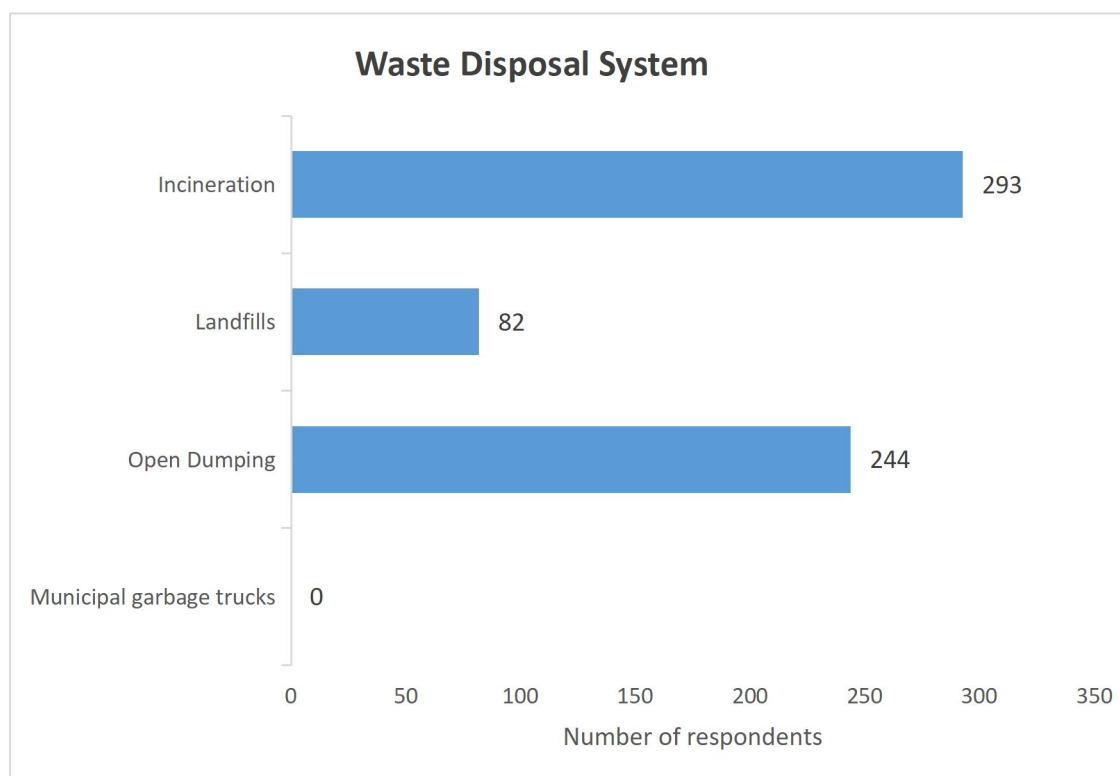


Figure 2.9: Waste disposal system available in the community..

2.7 Sewage Disposal Systems

Based on the data collected, incineration (293 respondents) emerges as the most prevalent method of waste disposal in Shao, closely followed by open dumping (244 respondents). Together, these two practices account for the vast majority of waste disposal methods used. Meanwhile, only 82 respondents utilize landfills, suggesting that proper landfill facilities may be limited or are not the preferred option for most households.

Notably, no respondents report using municipal garbage trucks, which indicates a lack of organized, government-provided waste collection services in the community. Consequently, most residents resort to self-managed disposal methods—either burning waste or dumping it openly—posing significant health and environmental risks. These findings underscore an urgent need for improved waste disposal systems, enhanced infrastructure, and heightened public awareness in Shao.

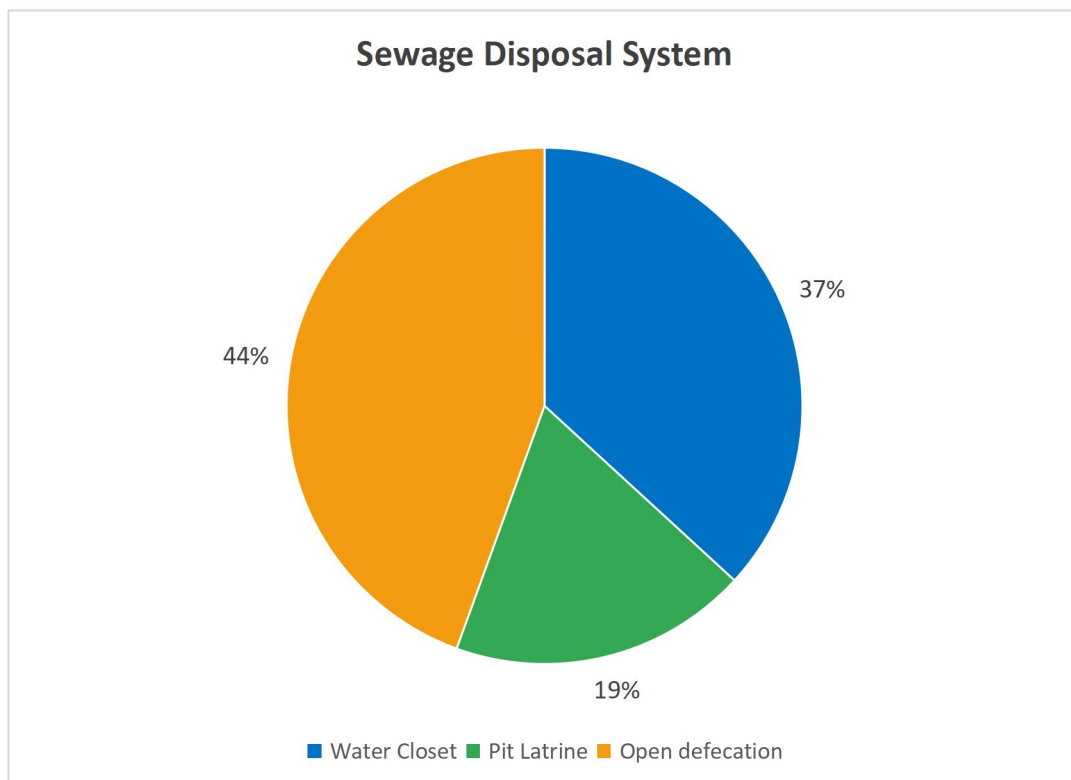


Figure 2.10: Sewage disposal system available in the community.

2.8 Prevalent Ailments in Shao

We enquired from both the community members and the health facilities present within the community about the ailments that were endemic in the community. The majority of the populace saw this as confidential information and some claimed to not be suffering from any illness. Others responded saying they were suffering from malaria, ulcer, hypertension, generalized pain, chronic cough, typhoid fever, eye defects, arthritis and other diseases.

From our enquiries at the primary healthcare center, we got similar information with malaria, ulcer and chronic cough being the predominant illness treated.

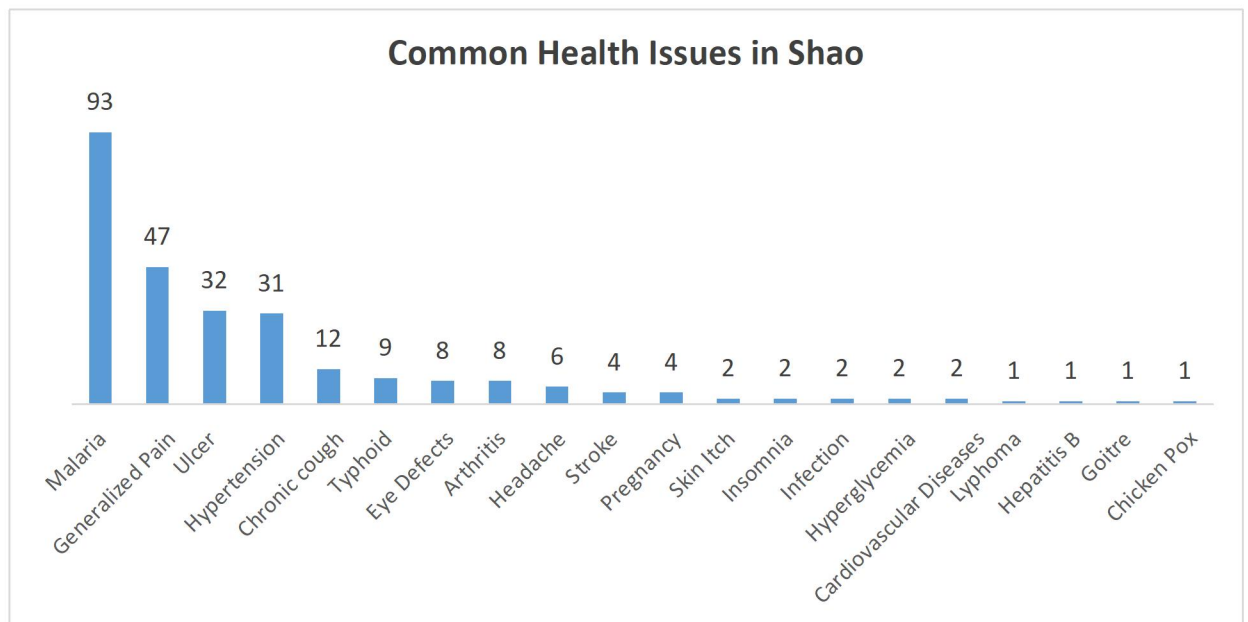


Figure 2.11: Prevailing health challenges in the community

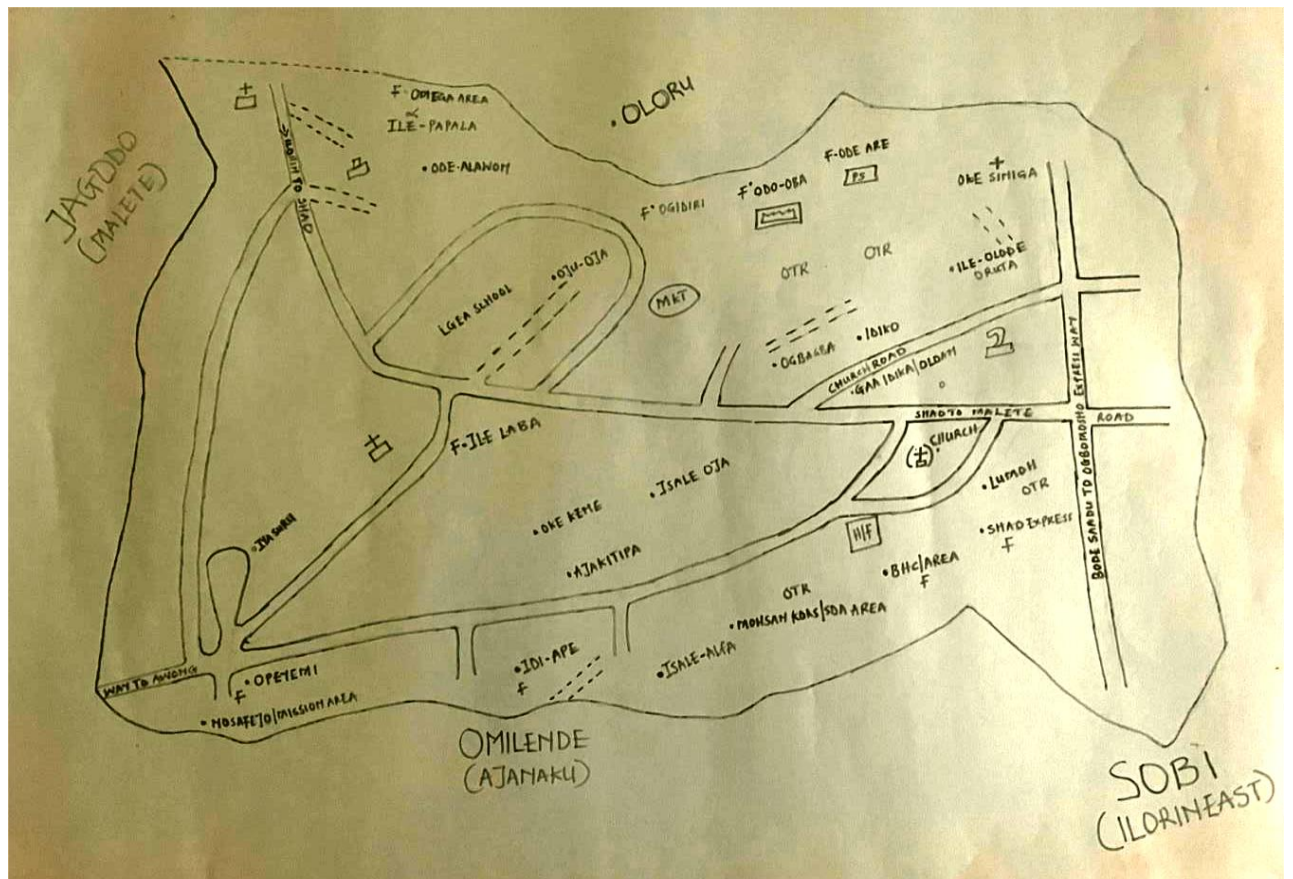


Figure 2.12: Map of Shao; Wards, Borders and Major Landmarks

MINIMUM LEARNING OBJECTIVE 1

TO EXTRACT DATA ON THE NUMBER AND TYPE OF HEALTH FACILITIES AND SERVICES PRESENT WITHIN THE COMMUNITY.

3.1 Introduction

To address our learning objective, a focused data extraction plan was developed to identify, categorize, and validate information on health facilities and services available within the Shao community.

3.2 Methodology

3.2.1 Definition and Scope

Our first step was to clearly define what constituted a “health facility” and the range of “services” they provided.

Health facilities and services refer to the physical structures, organizations, and professionals that provide healthcare services to individuals and communities. These include primary healthcare center (PHC), hospitals, clinics, laboratories, pharmacies, and chemist stores which offer preventive, curative, rehabilitative, and palliative care.

The king and the community leaders were briefed on the data extraction strategy and we received their approval to proceed with it. We organized 8 teams to gather targeted information for each ward in Shao to guarantee thorough coverage of the whole community.

On Wednesday, 19th of March, we set out to carry out the survey. In conducting this survey, we found out that some members of the community are involved in distribution and consumption of unorthodox health services and herbal medicines. We termed this category as “unlicensed practitioners”.

3.2.2 Tools and Process of Data Collection and Management

❖ Questionnaire Development and Orientation

- ✓ We designed a questionnaire using Google Forms, that included both closed- and open-ended questions.
- ✓ Key questions covered facility type (PHC, hospitals, clinics, laboratories, pharmacies, and chemist stores), service offerings (maternal health, general outpatient services, diagnostic services, pharmaceutical services, provision of over-the-counter drugs, etc.), and usage frequency by community members.
- ✓ Each of the eight teams received a brief orientation on administering the questionnaire.

❖ Field Administration

- ✓ We were divided into 8 teams for each of the ward in shao community. Team members accessed Google Forms using their mobile devices for in-person interviews where internet access was limited.
- ✓ In each ward, participants included both facility operators (for facility data) and community members (for usage data).

❖ Data Management and Analysis

- ✓ ***Data Entry:*** Responses were automatically collated in Google Forms, generating a unified spreadsheet of all responses.
- ✓ ***Data Cleaning:*** We reviewed entries for duplicates and inconsistencies (e.g., verifying the correct ward assignment). In cases of missing or ambiguous data, group leads revisited respondents for clarification.
- ✓ ***Excel-Based Analysis:*** We exported the cleaned data into Microsoft Excel. Using filters and pivot tables, we sorted facilities by type (PHC, clinic, chemist store, etc.)

and tallied the total counts. We similarly analyzed service usage, summarizing the number of community members who reported visiting each type of facility.

✓ ***Tabular and Graphical Outputs:*** The final Excel sheet reflects both the number and types of facilities and the level of usage by community members. We created simple bar charts or pie charts to illustrate usage distribution across wards.

3.3 Result Findings

From the data collected, the community has 1 primary healthcare center (PHC), 1 private clinic (named the Evangelical Church Wining All [ECWA] Misionary Clinic, 2 hospital-based laboratories (associated with the PHC and the ECWA clinic), 1 pharmacy (named Rainbow Healthcare Pharmacy), and 14 chemist stores. There are no private hospitals, secondary and tertiary hospitals. Out of 497 respondents, 229 respondents said they make use of the primary healthcare center, 174 respondents said they make use of the ECWA clinic, 196 respondents said they make use of the pharmacy and chemist stores, and 27 respondents said they make use of unlicensed practitioners such as those who make use of unorthodox services and herbal medicines.

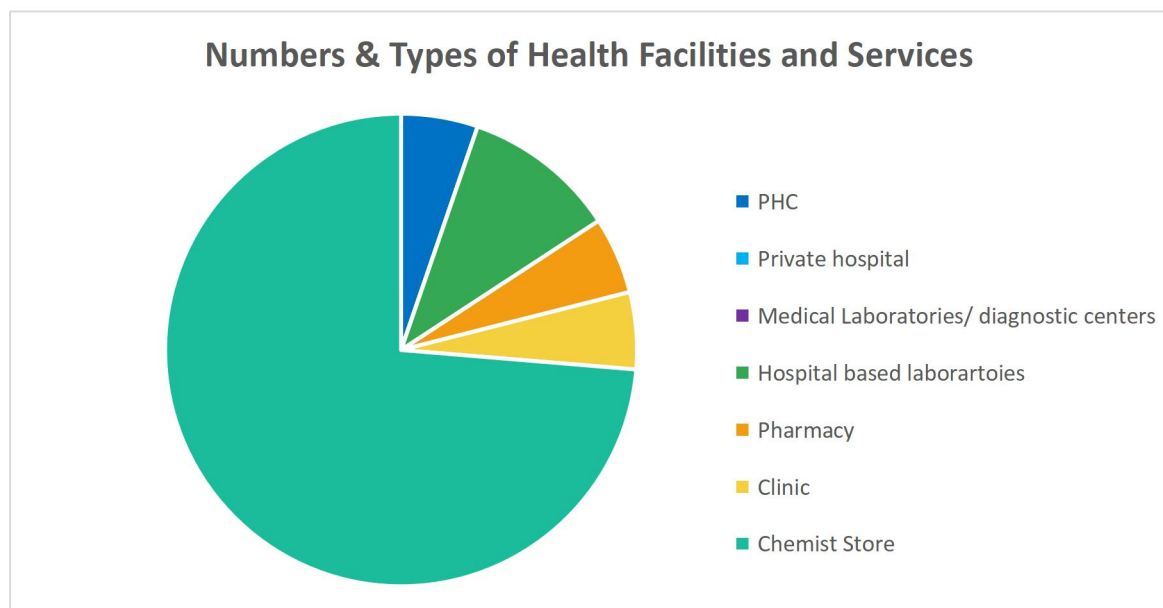


Figure 3.1: Pie Chart showing the numbers and types of health facilities and services

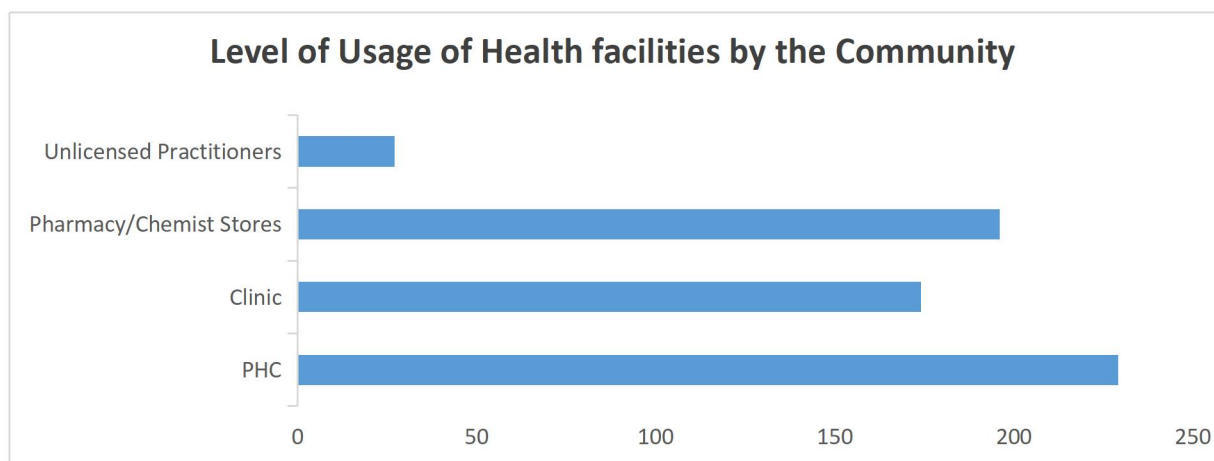


Figure 3.2: Bar Chart showing the level of usage of health facilities by the community

3.4 Detailed Facility Descriptions

3.4.1 The Primary Healthcare Center (PHC)

The PHC is the cornerstone of public health in developing regions, designed to provide accessible, affordable primary care in line with the Alma Mater Declaration of 1978. It is located at the center of Shao. The PHC employs a diverse team that includes:

- ❖ 2 Deputy Directors of Nursing Services (DDNS)
- ❖ 4 Community Health Extension Workers (CHEW)
- ❖ 1 Assistant Deputy Community Health Officer (ADCHO)
- ❖ 6 Health Attendants
- ❖ 4 Health Educators
- ❖ 1 Health Labourer
- ❖ 1 Laboratory Scientist
- ❖ 1 Laboratory Technician
- ❖ 1 Night Guard

The total staff strength is 21, and currently, there is no resident medical doctor.

The clinic days are:

- ❖ **Monday:** General Outpatient Department (GOPD) and daily immunization.
- ❖ **Tuesday:** Mass immunization.

❖ **Wednesday:** GOPD and daily immunization.

❖ **Thursday:** Ante-natal clinic.

❖ **Friday:** Geriatric clinic.

The services offered:

❖ **Immunization:** A comprehensive schedule that includes vaccines such as BCG, HepB, OPV, PENTA, Rota, IPV, Measles, Vitamin A, Yellow Fever, and Meningitis vaccines.

❖ **Emergency Care:** Provision of first aid and stabilization services before referring advanced cases.

❖ **Vaccination and Storage:** Equipped with a solar room and a cold room to maintain optimal vaccine storage conditions.

❖ **Sample Collection:** Facilities for collecting various samples (blood, urine, stool, saliva) for diagnostic purposes.

❖ **Delivery and Postpartum Care:** Includes a labour room and lying-in ward for maternal care and postnatal services.

❖ **Rapid Diagnostic Tests (RDTs):** Conducts quick tests for diseases such as malaria, hepatitis, and COVID-19.

❖ **Sensitization and Counselling:** Organizes outreach programs, health talks, and counseling sessions for community awareness and patient education.

❖ **Referral Services:** Refers patients to higher-level facilities such as University of Ilorin Teaching Hospital (UIH), Sobi Specialist Hospital when necessary.

❖ **Dispensation of Drugs:** Maintains a pharmacy that stocks basic medications and offers some tests and drugs free of charge.

❖ **Ante-Natal Care:** Provides essential maternal drugs and counseling at discounted prices.

- ❖ **PHC-Based Laboratory:** The PHC laboratory, staffed by 1 laboratory scientist and 1 laboratory technician, is a key component of the center. It supports sample collection and initial diagnostic testing. The facility ensures that samples are properly stored and, when needed, referred to more advanced diagnostic centers for further analysis. The laboratory is inadequately staffed with very limited diagnostic services being offered. Hematology and Microbiology services are very limited, while Histopathology and Chemical Pathology services are non-existent with patients being referred to laboratories outside the community. The Directly Observed Treatment, Short Course (DOTS) Laboratory is used to receive samples for tuberculosis which are referred to the UITH Medical Laboratory for testing.



Figure 3.3: Visit to PHC in Shao

3.4.2 The ECWA Clinic

It was established over 100 years ago and solely funded by the church. It serves as a critical health provider in the community. The clinic features three wards, a consultation room, an injection room, an outpatient department, and a laboratory equipped with fully functional equipment. It also includes a non-functional delivery room. It is managed by a Community Health Officer (CHO), and the clinical team includes one doctor, one nurse, and one laboratory technician. The services offered are:

- ❖ **Emergency Care:** First aid treatment for minor injuries and illnesses.
- ❖ **Dispensation of Drugs:** The on-site pharmacy dispenses exclusive drugs (e.g., ECWAquine syrup) produced at the ECWA pharmacy in Jos.
- ❖ **Maternity Services:** Used as a maternity center for delivering babies, offering paid but discounted services.
- ❖ **ECWA Laboratory:** The laboratory is currently not functioning.



Figure 3.4: Visit to ECWA clinic.

3.4.3 The Rainbow Healthcare Pharmacy

It is a privately owned pharmacy located at the junction of the street leading to the PHC. It was opened in February, 2023 with a team of four employees, including a certified pharmacist who is present at all times. The services offered include:

- ❖ Dispensation of drugs, primarily catering to diabetic and hypertensive patients (often with prescriptions from the PHC).
- ❖ Sales of drugs such as anti-malarials, antibiotics, analgesics, and blood tonics.
- ❖ Occasional prescription of medication when community residents present with complaints; however, it does not offer first aid or wound dressing services, referring patients to the PHC as necessary.

3.5 Limitations

- ❖ **Internet Connectivity:** Some wards had limited internet access, requiring offline form collection and subsequent online entry.
- ❖ **Data Consistency:** Multiple enumerators meant we had to ensure everyone applied consistent definitions (e.g., “clinic” vs. “unlicensed practitioner”).
- ❖ **Language Barrier:** Some community members spoke local dialects that our team was not fluent in, requiring translation assistance.
- ❖ **Restricted Access due to Topography:** Certain areas were difficult to reach, limiting our ability to survey all residents equally.
- ❖ **Unwillingness to Participate:** Some individuals were hesitant or skeptical about sharing health-related information, despite reassurances of confidentiality.

3.6 Conclusion

The community has a limited number of formal healthcare options, including one primary healthcare center, a private clinic, and several chemist stores. The significant involvement of community members with these facilities reflects an awareness of

healthcare services, though a considerable number of respondents also tend to seek care from unlicensed practitioners.

Despite the challenges encountered during the data collection—like internet connectivity problems, language difficulties, and participant reluctance—the methodology was largely effective in obtaining vital information. The results underscore the need to formalize and regulate health services while also considering the community's preference for alternative health practices.

Future initiatives should focus on improving the availability and accessibility of recognized healthcare services, and advancing health education. Collaborating with the community leaders and stakeholders will be essential for implementing strategies that genuinely enhance health outcomes for the Shao community.

MINIMUM LEARNING OBJECTIVE 2

TO DETERMINE THE VARIOUS CHALLENGES MITIGATING EASY ACCESS TO SATISFACTORY HEALTH SERVICES.

4.1 Introduction

Access to healthcare is a fundamental human right, recognized by the World Health Organization (WHO), yet numerous barriers often hinder individuals from receiving the care they need. This report will explore these challenges within the context of Shao, drawing upon observations, community interactions, and relevant literature.

4.2 Background

Access to healthcare, as defined by the WHO under the (AAAQ) framework includes the following:

- ❖ *Availability:* The presence of adequate health facilities, trained healthcare professionals, and essential medicines.
- ❖ *Accessibility:* The ability of individuals to physically reach health services, considering factors like distance, transportation, and cost.
- ❖ *Affordability:* The ability of individuals to pay for health services without incurring financial hardship.
- ❖ *Acceptability:* The cultural and social appropriateness of health services, considering factors like language, gender sensitivity, and traditional beliefs.
- ❖ *Quality:* The provision of safe, effective, and timely healthcare services that meet established standards.

However, when patients encounter barriers hindering their access to health facilities and services, it not only impedes their ability to receive necessary medical care but also undermines the establishment of meaningful relationships with healthcare providers, ultimately affecting overall patient satisfaction.

4.3 Approach

Our method for determining the challenges mitigating easy access to satisfactory health services in Shao involved conducting a community-based survey using an online questionnaire.

On Thursday, 20th of march, 2025, the group was divided into 8 teams consisting of about 8-9 individuals and set out for survey in the community using questionnaires to ask residents the reasons or factors that prevent them from easily accessing healthcare services. We interviewed a total amount of 497 people.

4.4 Tools and Process of Data Collection and Management

We used identical tool and process of data collection and management has established in preceeding chapter to organize information collected and to draw results.



Figure 4.1: Data collection

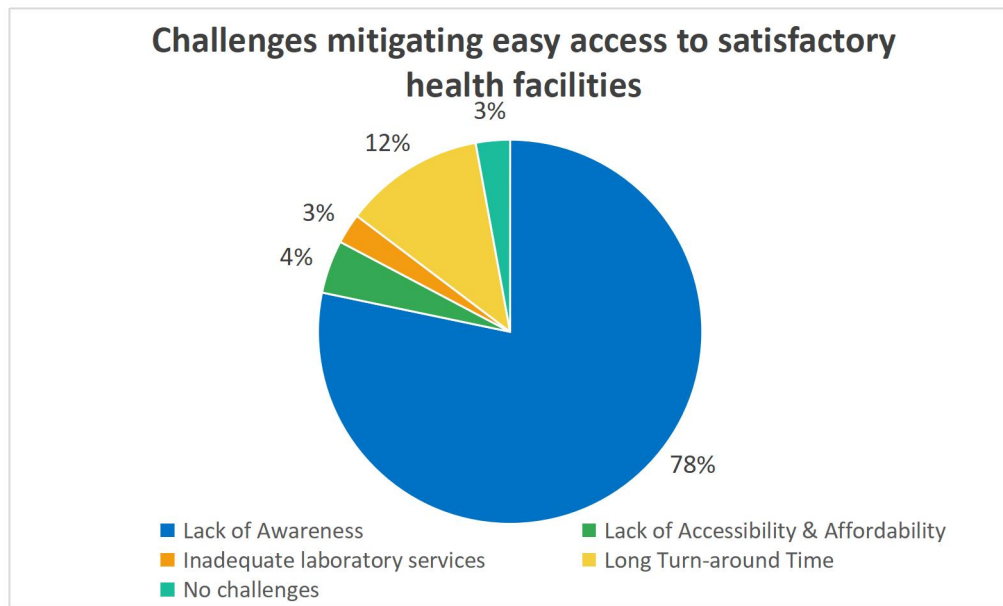


Figure 4.2: Pie chart showing the challenges mitigating against easy access to healthcare services.

4.5 Results of Findings

- ❖ **Lack of Awareness:** According to data collected, majority of the community are unaware of the available healthcare services, preventive measures and treatment options within the community due to certain factors such as traditional beliefs, myths and misconception about disease and modern medicine. Also illiteracy within the community makes it difficult for individuals to understand health-related information including when and where to seek medical help.
- ❖ **Lack of Infrastructure:** The analysis of the gathered data indicated that some community members are dissatisfied with healthcare facilities due to poor infrastructure, which affects their comfort. Specifically, both the PHC and the ECWA maternity clinic lack essential amenities like toilets, creating challenges for both patients and staff. This shortcoming not only causes discomfort for admitted patients but also disrupts laboratory procedures, as individuals may need to find alternative locations for sample collection. As a result, some residents prefer to seek medical care from facilities outside the community to access better services and infrastructure.

- ❖ **Lack of Accessibility and Affordability:** After the analysis of the data collected, some of the community members expressed their concerns about their inability to access the healthcare facilities due to factors such as distance barriers between their homes and healthcare facilities, poor infrastructure and shortages of professional healthcare personnel. This makes it difficult for them to adequate healthcare services. They also expressed their concerns of financial constraints in accessing healthcare services as a result of high cost of healthcare services and medication making it financially difficult to seek medical help.
- ❖ **Cultural and Religious Beliefs:** It was observed that the locals preferred traditional healing methods (herbs, popularly known as "agbo") or spiritual practice over conventional medicine. According to the health workers at the Primary health Centre, they identified this as a huge problem because this preference has led individuals to delay or avoid seeking healthcare from mainstream providers, hindering timely diagnosis and treatment of medical conditions. Some staff at ECWA Clinic said some religious groups have objections to certain medical interventions, blood transfusions, or organ donation. These beliefs pose challenges for healthcare providers in delivering comprehensive care that aligns with patients' religious values while also addressing their medical needs effectively.
- ❖ **Inadequate Laboratory Services:** According to the data collected, some of the members of the community state their difficulties in lack of sufficient and proper equipped laboratory facilities, personnel and resources needed to conduct medical tests and diagnostics effectively which results in delay in diagnosis, misdiagnosis and inadequate patients care. They stated that only PHC offers laboratory services in the whole community and sometimes when they're not properly diagnosed, they sort medical services from the nearest town.

- ❖ **Delayed Turn Around Time:** According to data collected, most of the members of the community shows concern for long turn around time when they sort healthcare services such as lengthy wait time for medical consultation, delay in laboratory testing and diagnosis, extended wait time for prescription or drugs. This hinder their ability to rely on healthcare services and sort alternative with the traditional health practitioners.

4.6 Recommendations

- ❖ **Development of Infrastructure and Supply Provision:** Collaboration between the government and non-governmental organizations (NGOs) is essential to ensure a steady supply of essential medications and the enhancement of healthcare infrastructure. Key amenities such as toilets, mattresses, mosquito nets, and other necessary supplies should be provided to improve patient comfort in medical facilities. These joint efforts will not only meet urgent needs but also play a significant role in enhancing overall healthcare delivery and patient satisfaction.
- ❖ **Boosting Awareness:** Implement community-based health education programs to inform residents about available medical services, disease prevention, and treatment methods. Utilizing local media channels such as radio, television, and social media, alongside community leaders, to effectively share health-related information. And also developing multilingual educational resources to accommodate diverse populations with varying literacy levels will be of great impact to boosting awareness.
- ❖ **Minimizing Long Turnaround Time:** Enhancing hospital management systems by incorporating digital record-keeping and automated appointment scheduling to reduce patient wait times. Expanding the healthcare workforce to improve service efficiency and accessibility. Also providing Modernize laboratory facilities with advanced equipment and technology to expedite diagnostic processes. Introducing a triage

system in hospitals to ensure that critical cases receive immediate attention within the community.

- ❖ **Enhancing Laboratory Services:** Upgrading laboratory facilities with advanced diagnostic equipment to improve the accuracy and efficiency of test results. Increasing the recruitment and training of laboratory technicians and specialists to mitigate workforce shortages. And also Implementing strict quality control protocols to ensure consistent and reliable testing outcomes can help enhance laboratory services.

MINIMUM LEARNING OBJECTIVE 3

TO DETERMINE THE IMPEDIMENT INHERENT WITH THE HEALTH SERVICES PROVIDERS AND FACILITIES MAKING ACCESS TO SATISFACTORY HEALTH CARE A DAUNTING TASK.

5.1 Introduction

Health care providers and facilities are the life line of any community, their absence or incompetence results in situations like high mortality rates, epidemic of infectious diseases, and a general decline in community health, hence cementing their importance as one of the pillars of any communities health. As one of the learning objectives undertaken by our group, we were saddled with the responsibility of identifying the impediments inherent in the health care service providers and facilities. This report summarizes our findings and highlights the major problems faced by the community in accessing satisfactory healthcare.

5.2 Approach

Our entire class was subdivided into smaller groups embodying a divide and conquer approach and each group was responsible for a particular geographic location within the shao community. Each subgroup was tasked with retrieving relevant information towards the objective and they were equipped with questionnaires to guide the manner of inquiries, these subgroups visited public health facilities ranging from roadside chemists all the way to primary health care center. The mode of delivery of services within these facilities was observed in addition with interviews of stakeholders in line with the questionnaires. Our assessments can be categorized into 2 sections:

- ❖ The healthcare service providers
- ❖ The healthcare facilities

5.2.1 The Healthcare Service Providers

- ❖ **Insufficient personnels:** Using the laboratory as a case study, the scientist responsible for the laboratory at the primary healthcare center within the shao community is responsible not only for that PHC but for all the PHC in all the 16 local government in Kwara state. These is a testament to the constant cry of insufficient medical personnel from doctors to nurses to medical laboratory scientist plaguing the Nigerian healthcare system.
- ❖ **Language Barrier:** It was observed during our visit to the PHC that all the medical personnel were Yoruba speakers. Shao is a community containing a number of ethnic groups predominantly Yoruba and Hausa. Some of the staff at the PHC emphasized on the problems faced due to language barriers which have created many challenges that have made access to satisfactory healthcare problematic for many individuals.

5.2.2 The Healthcare Facilities

- ❖ **Poor infrastructure:** According to the data collected though the survey that was conducted, we were able to determine that of the seventeen healthcare facilities present in the community, none have adequate and functional infrastructure, 3 had adequate but non-functional infrastructure, 13 had inadequate but functional infrastructure and 1 of them had inadequate and non-functional infrastructure. The healthcare facilities in the community are inadequate and non standard,lacking essential equipment and amenities. This hinders the delivery of quality healthcare services, compromising patient care and safety.
- ❖ **Lack of electricity:** Most of the community lacks a reliable and stable electricity supply with only 6 healthcare centers having alternate power supply like solar inverter and generators. This significantly hinders the effective running of the healthcare facilities and reduces the quality of living of members of the community. This

limitation affects the storage of vaccines, medications, blood for transfusions, patient samples and reagents, as well as the use of essential medical equipment.

5.3 Recommendations

❖ Strengthening Healthcare Service Providers Increase and Decentralize Staffing:

- ✓ *Recruitment & Deployment:* The state should collaborate with the Federal Ministry of Health and training institutions to recruit additional doctors, nurses, laboratory scientists, and other healthcare workers. Establishing dedicated personnel for each PHC (rather than one lab scientist covering 16 LGAs) would help improve service delivery.
- ✓ *Retention Incentives:* Develop retention strategies such as competitive remuneration, rural hardship allowances, and career development opportunities to discourage brain drain and motivate existing staff.

❖ Enhance Language Competency and Cultural Sensitivity:

- ✓ *Multilingual Training:* Organize periodic language and cultural sensitivity training to equip healthcare workers with basic communication skills in Hausa (or other prevalent local languages). This could include refresher courses or hiring bilingual interpreters.
- ✓ *Recruit Diverse Staff:* Encourage recruitment from a wider pool of candidates, including those from different ethnic backgrounds, to reflect the community's diversity and reduce language barriers.

❖ Adopt Telemedicine and Task-Shifting Strategies:

- ✓ *Remote Support:* Utilize telemedicine solutions to provide specialist advice and reduce the workload on limited personnel. This approach can also serve as a tool for continuous professional development.

- ✓ *Task Shifting*: Empower trained community health workers to support routine tasks (e.g., basic diagnostics, patient follow-up), freeing up specialized staff to focus on more complex care needs.
- ❖ **Improving Healthcare Facilities Infrastructure Renovation and Standardization:**
 - ✓ *Facility Upgrades*: Secure funding from state and federal sources, as well as explore public–private partnerships, to renovate existing PHCs. This includes structural repairs, ensuring proper sanitation, water supply, and the provision of essential amenities.
 - ✓ *Equipment Procurement & Maintenance*: Invest in modern, standardized medical equipment. Establish regular maintenance schedules and inventory audits to ensure that essential devices remain operational.
- ❖ **Reliable Electricity Supply:**
 - ✓ *Renewable Energy Solutions*: Implement off-grid solar energy systems with battery storage to ensure continuous power supply. Such systems are proven to enhance healthcare delivery by supporting refrigeration for vaccines, running essential medical equipment, and maintaining overall facility operations.
- ❖ **Capacity Building and Regular Supervision:**
 - ✓ *Training on Facility Management*: Organize workshops for facility managers and staff on best practices for maintenance, resource management, and emergency preparedness.
 - ✓ *Monitoring and Evaluation*: Set up a supervisory framework with periodic assessments of facility infrastructure, equipment functionality, and service quality. Data-driven audits can help in timely identification of shortcomings and guide resource allocation.

By simultaneously addressing both human resource constraints and facility deficiencies, these recommendations aim to enhance overall healthcare delivery, improve patient safety, and ensure that diverse community needs are met effectively.

MINIMUM LEARNING OBJECTIVE 4

TO ASSESS THE LEVEL OF GOVERNMENT AND PRIVATE SECTOR INVOLVEMENT IN HEALTH PROVISION IN THE COMMUNITY

6.1 Introduction

Assessing the level of government and private sector involvement in health provision in the community involves examining the various initiatives, establishments, and efforts made by government entities (local, state, or federal) and private sectors, including religious organizations, schools, Non-Governmental Organizations (NGOs), and individual firms towards providing easy, accessible, and affordable healthcare services to the community at large. Therefore, this assessment aims to evaluate the extent of their involvement, identify areas of strength and weakness, and provide recommendations for improvement.

6.2 Methodology

As part of our community posting, we conducted a demographic survey in various parts of the Shao community on 20th March, 2025 and one of the aims of the survey is to gather insights into the level of commitment from both the government and private sector in providing accessible and affordable healthcare in the community. This Minimum Learning Objective (MLO) presents our findings and recommendations for enhancing healthcare outcomes in the Shao community. We aimed to:

- ❖ Identify existing healthcare facilities, services, and initiatives put in place by the government and private sectors.
- ❖ Evaluate community members' awareness of government and private sector investments and their quality, accessibility, and affordability.
- ❖ Identify the challenges limiting community members from maximum utilization of available and established government and private sector investments.

- ❖ Identify areas for improvement in healthcare provision and community engagement.

6.3 Results of Findings

6.3.1 *Government Involvement*

- ❖ Establishment of a well-structured Primary Health Care (PHC) facility in the community.
- ❖ Construction of a public toilet at the major market in the community by the Federal Ministry of Water Resources.

6.3.2 *Private Sector Involvement*

- ❖ **The ECWA Clinic:** It is a private religious health care system put in place for the sole purpose of delivering health care at a subsidized rate to the members of the community. This clinic has contributed in no small way to the community as many of the members of the community attest to not just its availability but to their patronage and its affordability. The ECWA Clinic gets funds and sponsorship through **the Sons and Daughters of the Soil** involvement. According to a health care worker at the clinic, the sons and daughters meet at a specific time of the year and make contributions in one way or the other towards the advancement of the health care services at the Clinic. Noteworthy is that a lot of the drugs used in this clinic are produced and sponsored by the ECWA pharmaceutical company located in Jos, Nigeria.
- ❖ **The Rainbow Pharmacy:** The only major and private pharmacy in Shao town also contributes to the health of the community members by providing easy and timely access to standardized drugs, providing ease in bodily functions check-ups in order to monitor the state of health, thus, preventing complications.
- ❖ **The Wellbeing Foundation Africa; the Liverpool School of Tropical Medicine; Johnson and Johnson:** These are all private organizations involved in the

improvement of the availability and quality of newborn care in Kwara State. The Primary Health Care Centre in Shao is one of the beneficiaries.

- ❖ **The Hygeia Foundation in collaboration with the National Agency for the Control of AIDS and The Global Fund:** these organizations support the Primary Health Care Centre in Shao in a bid to fight HIV/AIDS and other diseases such as Tuberculosis and Malaria, they provide comprehensive HIV management, HIV/TB collaborative activities and malaria treatment.

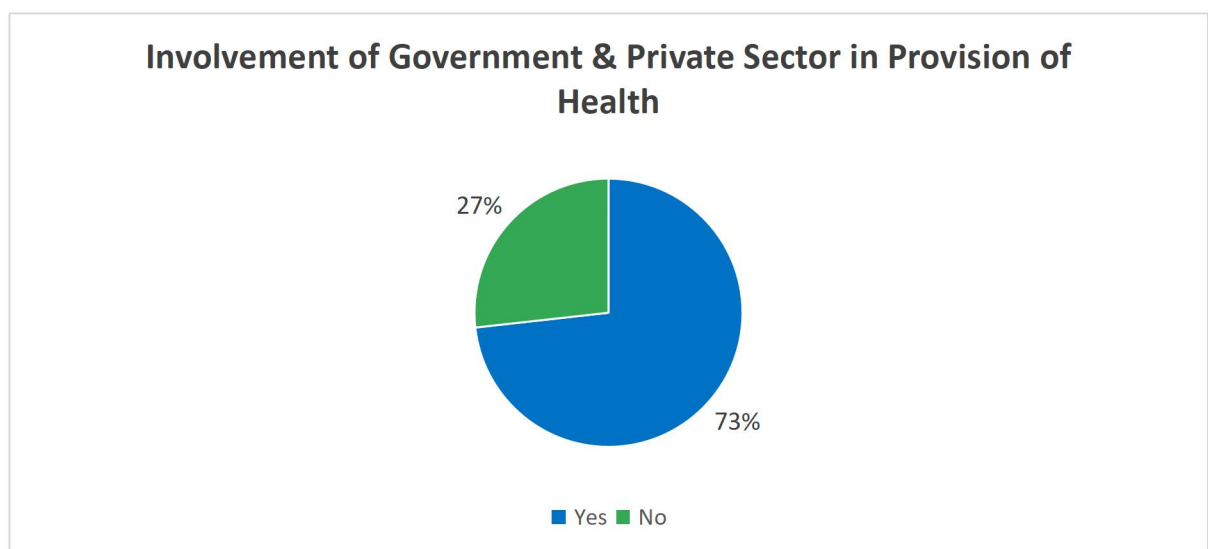


Figure 6.1: Involvement of Government & Private Sector in provision of health

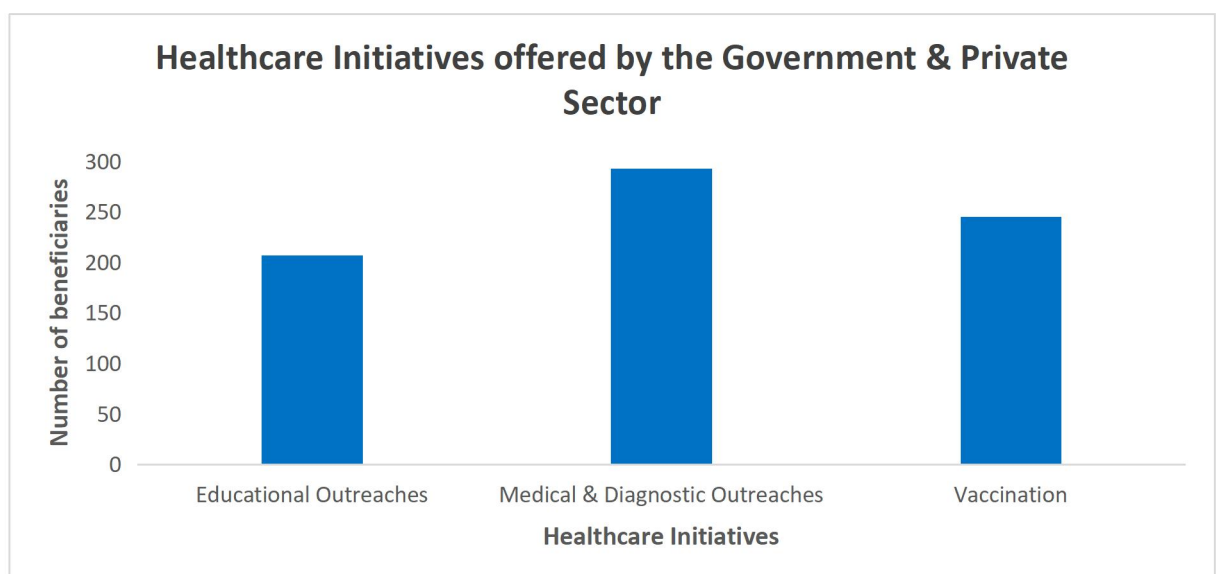


Figure 6.2: Healthcare initiatives offered by the Government & Private Sector

MINIMUM LEARNING OBJECTIVE 5

TO SENSITIZE THE PEOPLE AND COMMUNITY ON THE IMPORTANCE AND PROACTIVE MEASURES TO FACILITATE HEALTH PROMOTION

7.1 Introduction

Dean Ornish once said “Awareness is the first step in healing.” This highlights the importance of sensitization which equips the people and community with the right knowledge that will not only prevent diseases but improve the overall health well being of the populace. Creating awareness is therefore a necessary step to promote and publicize, the proactive measures that improve health.

7.2 Health Sensitization

During our stay in the community of Shao, we had different opportunities which we seized to sensitize the people and community which included:

- ❖ Religious outreach
- ❖ School outreach
- ❖ Community Sensitization Rally

7.2.1 Religious Outreach

On 23rd March, 2025, we proceeded to one of the churches “First Baptist Church.” We had previously informed the church on our plans to sensitize the congregation and so after arriving we attended the church service and towards the end we were given the time to share the information we had. We conducted the sensitization program using Yoruba, the major native language spoken in the area. Our focus was on tuberculosis (TB), where we discussed its symptoms, modes of transmission, prevention strategies, and the importance of seeking medical attention and laboratory testing.

To enhance understanding, we carried placards with visual illustrations depicting tuberculosis symptoms and transmission. This approach helped reinforce key messages, ensuring that even those with limited literacy could grasp the essential information.

Additionally, we emphasized the importance of building immunity, especially among the elderly, by consuming fresh fruits, vegetables, and maintaining a well-balanced diet. This aligns with the World Health Day 2025 theme, “Healthy Beginnings, Hopeful Futures,” which underscores the need for proactive health measures to ensure long-term well-being and strengthening immunity is particularly vital in preventing infections like tuberculosis and other diseases prevalent in the community.

The engagement was interactive, as we encouraged participants to ask questions, fostering a deeper understanding of tuberculosis . This participatory approach not only increased awareness but also empowered community members to take charge of their health.



Figure 7.1: Religious outreach

7.2.2 School Outreach



Figure 7.2: School outreach

On March 24th, 2025, we carried out a health sensitization program across secondary schools in Shao community.

- ❖ Muslim Community Senior Secondary School
- ❖ Muslim Community Junior Secondary School
- ❖ ECWA Senior Secondary School
- ❖ ECWA Junior Secondary School
- ❖ 7th Day Adventist Secondary School

Our goal was to educate students on key public health concerns: tapeworm, hookworm, malaria, and tuberculosis, diseases that significantly impact communities like Shao, where open defecation and other environmental factors contribute to endemic infections.

We divided ourselves into five groups, with each group assigned to a different school. Two groups focused on tuberculosis awareness, in commemoration of World Tuberculosis Day, while the remaining groups covered malaria and intestinal parasites. At each school, we engaged the students in interactive sessions, explaining the general scope, symptoms, transmission, and prevention of these diseases. We also aimed to enhance understanding, by using visual illustrations depicting symptoms, transmission routes, and the life cycles of parasites like hookworm and tapeworm. These visual aids helped reinforce key messages, making the sessions more engaging and effective.

Beyond raising awareness, we aimed to inspire behavioral change among the students by fostering an interactive discussion. While we primarily focused on educating them, we hope that the knowledge gained will not only influence their daily lives but also be shared with their families and peers, contributing to wider community health improvement.

This initiative aligns with the World Health Day 2025 theme, “Healthy Beginnings, Hopeful Futures,” which emphasizes proactive health education and preventive measures to create a healthier future for communities. By educating young minds, we hope to instill lifelong habits that promote disease prevention, hygiene, and overall well-being in Shao.

7.2.3 Community Sensitization Rally

Sensitization does not end until every individual is informed. With this commitment in mind, on March 25th, 2025, we took to the streets of Shao community to raise awareness about one of the most crucial proactive steps in health promotion: getting tested before treatment.

Using a public address system, we conducted a moving rally, educating the public on the importance of laboratory testing before taking medication. We emphasized that diagnostic testing is essential not only for effective treatment but also for early detection of

silent ailments such as hepatitis, diabetes, and high blood pressure, as well as infectious diseases like malaria and tuberculosis. Beyond the rally, we engaged with community members individually, visiting their shops and homes to reinforce our message on the significance of proper diagnosis. To further drive awareness, we carried placards with impactful messages such as:

- ❖ “No Test, No Diagnosis.”
- ❖ “Quality Laboratory Services, Equals Quality Healthcare.”
- ❖ “Ile Ayewo, Eto Yin Ni.”

Additionally, we used this opportunity to invite community members to the health outreach scheduled for the following day, where various free diagnostic tests were conducted as part of our broader effort in health promotion and disease prevention.

This achieved the following minimum learning objectives:

- ❖ To carryout sensitization and advocacy on the importance of accessing medical laboratory services
- ❖ To emphasize on the right of patient to access effective medical laboratory services before treatment is administered.



Figure 7.3 Community Sensitization Rally

7.3 Community Outreach

On March 26th, 2025, we conducted a community health outreach, strategically located at the Oba's palace to ensure maximum participation and accessibility for the people of the community. The tents, chairs, tables and equipment were set up as early as 8:30am and the program began by 9am. The turnout was incredible, with 97 individuals participating in the free diagnostic tests.

7.3.1 Free Testing

To ensure a smooth workflow during the free diagnostic tests, we organized ourselves into different sections:

- ❖ Registration unit
- ❖ High blood pressure screening point
- ❖ Height and weight measurement point (BMI)
- ❖ Phlebotomy unit
- ❖ Malaria testing unit
- ❖ Tuberculosis screening unit
- ❖ Hepatitis B testing unit
- ❖ Urinalysis point

This structure allowed for an efficient testing process. Additionally, we set up a single tent as a waiting area where individuals could stay before testing and collect their results afterward. The Medical Laboratory Scientist from the Primary Health Care Center was also in attendance, whom we invited to sign out results before participants were discharged.

The free tests we conducted to serve as practical examples for members of the community, to ensure that they realize the need for regular health check ups and to promote early detection and proper disease management included:

- ❖ Malaria screening using Rapid Diagnostic Test (RDT) kits

- ❖ Tuberculosis screening
- ❖ Hepatitis testing
- ❖ Urinalysis
- ❖ Blood pressure checks for hypertension detection

7.3.2 Sensitization

Beyond providing these essential tests, we continued our sensitization efforts using a public address system, emphasizing the importance of hospital visits and proper testing for accurate diagnosis and effective treatment. We reinforced the message that testing before treatment is a proactive step toward better health outcomes and disease prevention.



Figure 7.4: Free testing and sensitization.

7.3.3 Results Of Tests

The data collected from the community diagnostic outreach provides key insights into the demographic distribution, health status, and prevalence of certain conditions among the participants.

❖ Demographics and Population Distribution

The majority of participants (48%) fall under the "Adults (no age info)" category, making it difficult to assess precise age-based health trends. Among those with recorded ages, most participants belong to the 50–59 and 60–69 age groups. Gender distribution is heavily skewed, with 78% of participants being female and only 22% male.

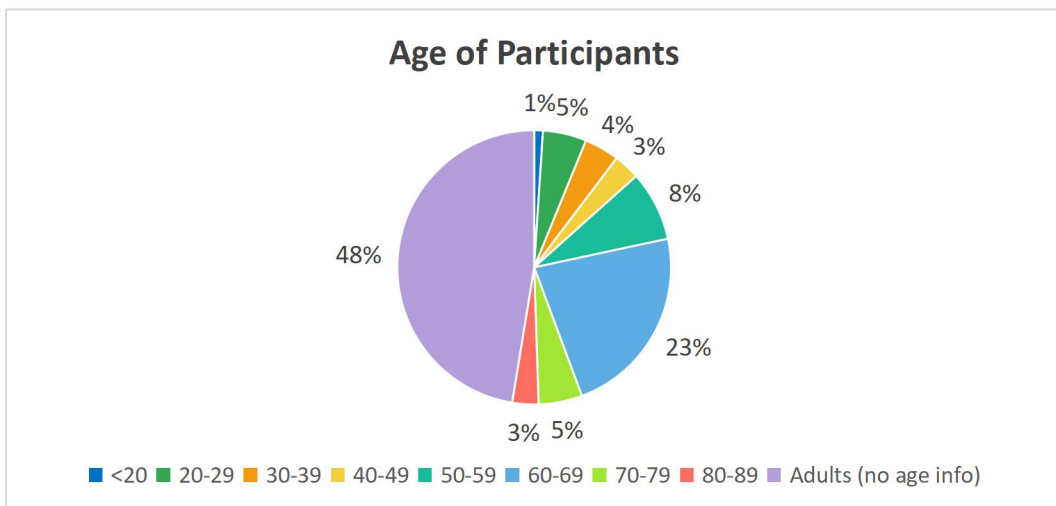


Figure 7.5: Age distribution of participants

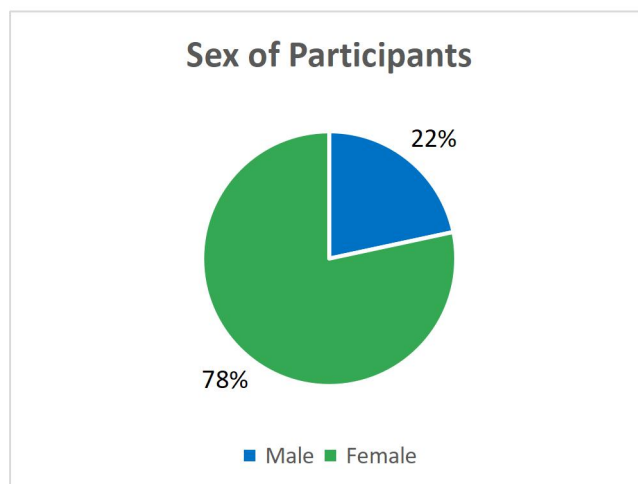


Figure 7.6: Sex distribution of participants

❖ Body Mass Index (BMI) and Weight-related Health Risks

A significant portion (38%) of participants are overweight, while 25% fall under different obesity classes (Obesity Class I, II, and III combined). Only 36% of the participants have a healthy weight, and a small fraction (1%) are underweight. The high prevalence of overweight and obesity indicates a potential risk for metabolic disorders such as diabetes and cardiovascular diseases.

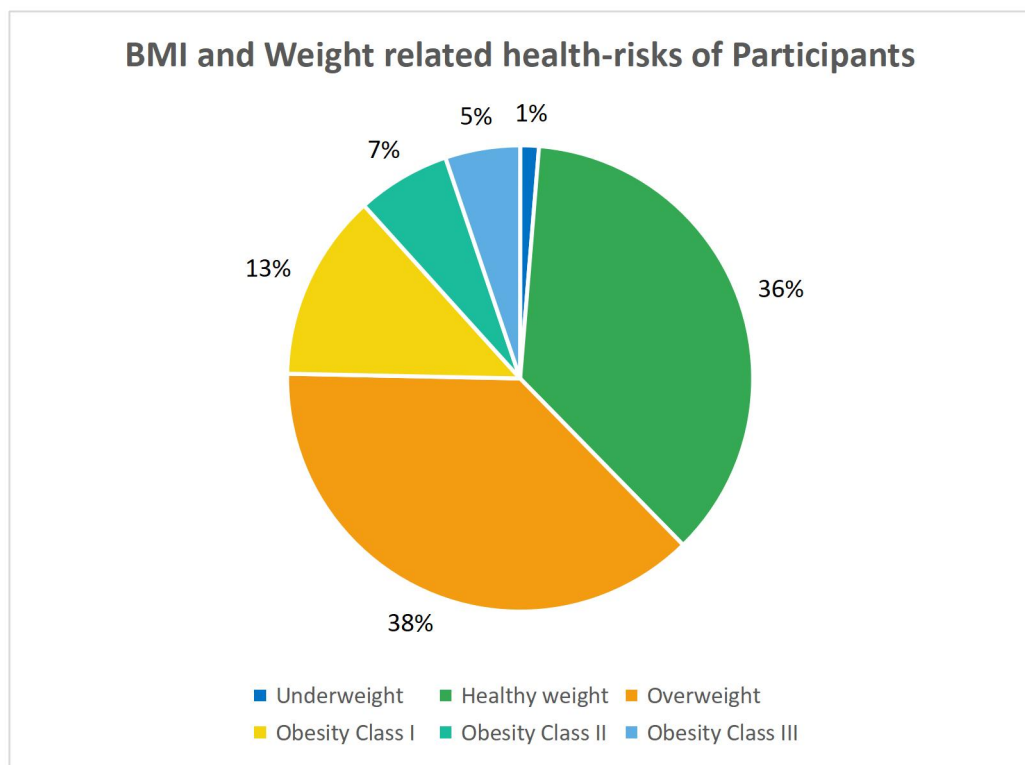


Figure 7.7: BMI and weight related risk of participants

❖ Blood Pressure and Hypertension Prevalence

Normal blood pressure was recorded in only 31% of the participants. 10% of individuals had elevated blood pressure, indicating early signs of hypertension. Hypertension was prevalent, with: 16% having Hypertension Stage 1, 33% having Hypertension Stage 2, and 10% experiencing Hypertensive Crisis (>180/120mmHg), which is a critical condition requiring urgent medical intervention. The high prevalence of hypertension suggests the need for better lifestyle management, dietary changes, and possibly medical intervention within the community.

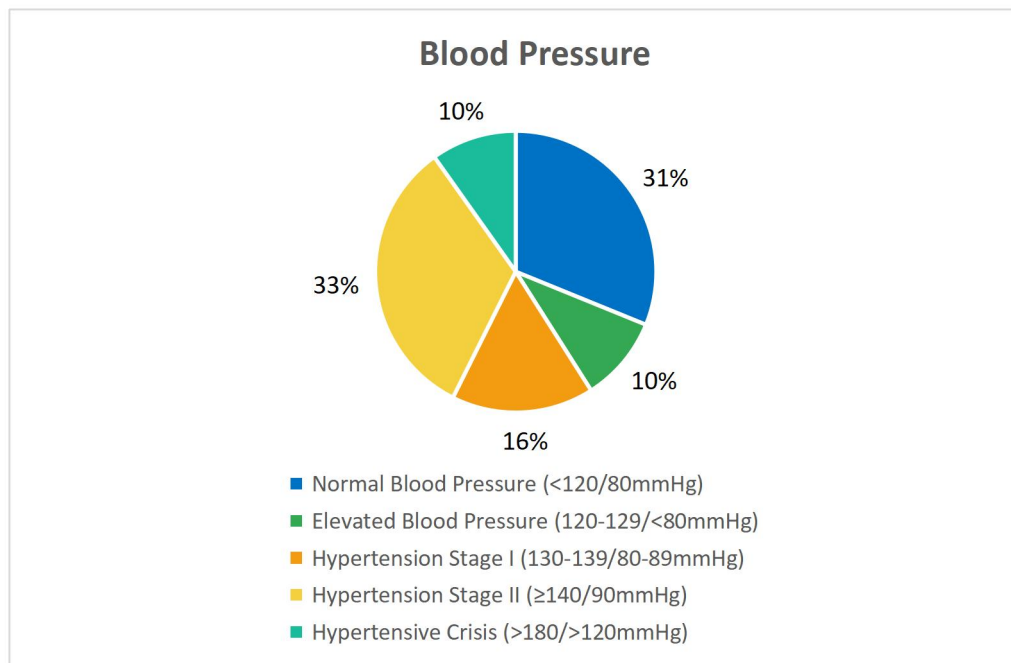


Figure 7.8: Blood pressure of participants

❖ Urinalysis Results and Possible Renal Issues

Protein was found in the urine of 11 participants, which may indicate kidney dysfunction or early signs of renal disease. Leucocytes (indicative of possible infections) were present in 12 cases. Ketonuria was detected in 7 participants, which may suggest diabetes, starvation, or other metabolic conditions. Other abnormal findings included the presence of blood in 3 cases and glucose in 1 case, potentially pointing to underlying diabetes or kidney-related disorders. The overall results suggest a need for further investigation into kidney health and metabolic diseases among the participants.

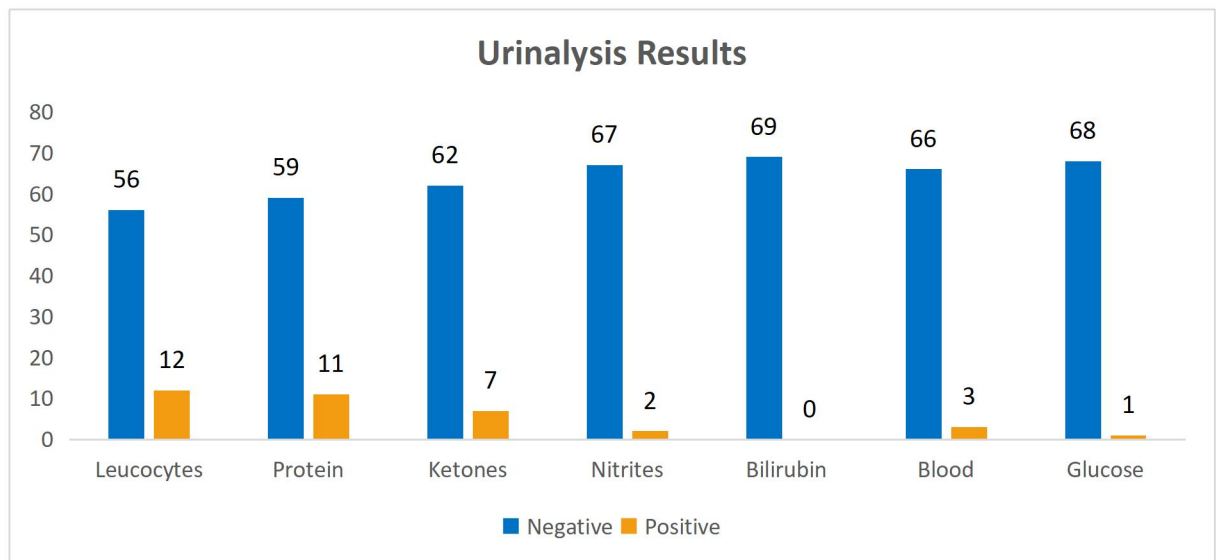


Figure 7.9: Urinalysis results of participants

❖ **Infectious Disease Screening**

- ✓ **Tuberculosis:** 1 case was reactive, while 31 were non-reactive.
- ✓ **Malaria (RDT):** 4 participants tested positive, while 36 were negative.
- ✓ **Hepatitis B (HBsAg):** 4 cases were reactive, while 47 were non-reactive.

Although the number of positive cases is relatively low, the presence of tuberculosis, malaria, and hepatitis B within the community highlights the need for continued surveillance, preventive measures, and possible treatment interventions.

7.3.4 Feedback Interviews

We also interviewed some of the participants about their experience during the outreach, and they all shared positive feedback, expressing gratitude and offering prayers.

7.3.5 General Implications and Recommendations

- ❖ **Hypertension and Obesity:** The high percentage of overweight and obese individuals, combined with the significant hypertension prevalence, indicates a need for awareness campaigns on lifestyle modification, diet, and exercise.
- ❖ **Kidney Health Monitoring:** The detection of proteinuria and abnormal urinalysis results suggests that screening for renal disease should be prioritized, particularly for those with hypertension or diabetes risks.
- ❖ **Infectious Disease Control:** Though relatively low, the presence of TB, malaria, and hepatitis B cases emphasizes the importance of vaccination, health education, and prompt medical care to prevent further spread.
- ❖ **Follow-up and Treatment:** Community health programs should consider providing follow-up services for individuals with abnormal findings to ensure they receive proper medical evaluation and management.

This outreach provides crucial insights into the community's health profile and highlights the importance of continued medical interventions, lifestyle education, and preventive healthcare strategies.

7.4 Conclusion

Our series of health sensitization programs and community outreach activities were driven by the goal of improving public health awareness and encouraging preventive healthcare practices which we achieve by educating students, engaging community members in discussions, and offering free medical tests, we aimed to instill a culture of proactive health-seeking behavior. While challenges remain, we hope that the knowledge shared will empower individuals to make informed health decisions and contribute to a healthier community.

MINIMUM LEARNING OBJECTIVE 6

TO MOBILIZE COMMUNITY RESOURCES IN PROVISION OF AT LEAST ONE PROJECT THAT WILL ENHANCE: “IT IS TIME TO BUILD A FAIRER. HEALTHIER WORLD FOR EVERYONE, EVERYWHERE”

8.1 Introduction

Shao is a hilly community in Kwara State, Nigeria, known for its mountainous terrain and traditional lifestyle. Through house-to-house surveys, physical observation, and interactions with residents and healthcare workers, our team identified several health-related challenges in the community. These include:

- ❖ Open defecation and indiscriminate dumping of waste, particularly on hillsides.
- ❖ Water scarcity caused by faulty boreholes and unreliable electricity.
- ❖ High prevalence of malaria and typhoid, linked to poor sanitation and consumption of unsafe water.
- ❖ Limited diagnostic capacity at the Primary Health Centre (PHC), especially for malaria and hematology services.

The following projects were chosen based on factors such as cost-effectiveness, relevance to public health, feasibility, and sustainability.

8.2 Community-Based Projects

❖ **Construction of an Incinerator:**

Due to widespread open dumping and open defecation observed during our surveys, our team constructed an incinerator to serve as a sanitary and safer method of waste disposal. This was selected after failed efforts to reopen an abandoned public toilet at Isale Oja due to potential political conflict, as it was previously built by a politician. This project aims to reduce environmental contamination and control the spread of vector-borne diseases such as typhoid and malaria.



Figure 8.1 Construction of the incinerator



Figure 8.2: Borehole Renovation



Figure 8.3: Provision of generator.

❖ **Borehole Renovation and Provision of Generator:**

Water scarcity was a prominent issue in Shao. The community relies mainly on boreholes, many of which are dysfunctional. Our group initially intended to repair manual boreholes at Oke Boo and the COBES unit site, but technical assessments revealed high costs and uncertainty regarding damage. We then identified a borehole located at Oke Sinniga/Isale Oja, originally donated by a youth corper. Its primary problem was inadequate electricity supply. We intervened by:

- ❖ Cleaning and washing the borehole water tank (untouched for over two years).
- ❖ Renovating the borehole environment.
- ❖ Donating a generator to power the borehole.

To ensure sustainability and proper monitoring, the generator was entrusted to a woman who owns a pepper-grinding shop beside the borehole and the P.A to the Ohoro of Shao, who lives directly behind the borehole. We proposed a sustainability plan that includes weekly or monthly contributions by residents of the community to cover fueling and routine maintenance of the generator.

This intervention improved water access for the community, especially in areas with near-total electricity blackout.

8.3 Laboratory/Diagnostic-Based Projects

❖ **Renovation of Laboratory Working Area at PHC Shao**

We observed unsanitary conditions in the PHC laboratory, including stained and damaged work surfaces. We hired a carpenter who replaced the laboratory bench to improve biosafety and infection control practices.

❖ **Facilitation of Microscopic Diagnosis of Malaria**

The scientist in charge of the lab disclosed that malaria testing was conducted only via Rapid Diagnostic Tests (RDTs), which are known to yield false positives due to cross-

reacting antibodies from other infections. The lab lacked staining materials for microscopy, despite owning a functional microscope donated by Hygeia Foundation. We provided Giemsa and Leishman stains for accurate microscopic malaria diagnosis. This upgrade enabled the PHC to adopt WHO-recommended diagnostic procedures for malaria

❖ **Provision of Hematocrit Reader**

We also responded to the need for packed cell volume (PCV) testing by donating a hematocrit reader to the laboratory. This tool is vital in evaluating anemia and other hematological conditions.



Figure 8.4: Renovation of laboratory in the PHC

8.4 Recommendations

For the Community and Local Leaders:

- ❖ Establish a community-led committee to oversee maintenance of the incinerator and borehole facilities.
- ❖ Implement and enforce proper waste management practices through education and community meetings.
- ❖ Ensure the proposed generator maintenance plan is formalized with transparent collection and spending of contributions.

For the PHC:

- ❖ The PHC requires increased government intervention and consistent funding, as most of their equipment, including microscopes, were donated by NGOs.
- ❖ Implement a reliable supply chain system for reagents and essential diagnostic tools.
- ❖ Provide continuous training for laboratory staff on standard diagnostic protocols and biosafety measures.

For the University and COBES Coordinators:

- ❖ Consider providing small grants to support community project implementation.
- ❖ Encourage continuity by having new student teams build on previous projects.
- ❖ Facilitate engagement with local artisans and health officials before COBES postings to smoothen planning.

8.5 Conclusion

Through MLO 6, we executed both community and diagnostic-based projects that addressed pressing public health challenges in Shao. We were able to:

- ❖ Improve environmental sanitation through incinerator construction.
- ❖ Enhance access to clean water via borehole rehabilitation.

- ❖ Boost diagnostic capacity through laboratory renovation, provision of reagents, and essential equipment.

These interventions align with the spirit of WHO's 2025 goal of achieving “Health Beginnings, Hopeful Futures” by providing the community with improved access to sanitation, clean water, and quality diagnostics.

MEDICAL LABORATORY SCIENCE SPECIFIC MINIMUM LEARNING OBJECTIVES (MLS MLOs)

9.1 MLS MLO 1

*TO IDENTIFY THE LEVEL OF ACCESS TO MEDICAL LABORATORY SERVICES IN
THE COMMUNITY*

9.1.1 Introduction

Medical laboratory services which include the analysis of patients samples such as body fluids, blood, tissues which are necessary in the accurate diagnosis, monitoring and treatment of diseases. Medical laboratory science plays a pivotal role in healthcare by providing essential diagnostic services that inform early medical interventions and preventive care. Accurate and timely laboratory testing is fundamental in identifying health conditions early, guiding treatment decisions, and monitoring disease progression, all of which contribute to improved health outcomes.

The integration of robust medical laboratory services into healthcare systems, especially in underserved areas, can enhance disease prevention, facilitate early detection of health issues, and promote equitable access to quality healthcare and even prevent epidemics. Hence, the reasons why it became paramount that we identified the level of access to medical laboratory services in Shao community

9.1.2 Methodology

To gather this information, on the 19th of March, 2025, we divided ourselves into eight groups so as to cover the eight divisions of Shao and with the aid of the prepared questionnaire and interactions with the people we were able to identify the level of access to medical laboratory services in the community.

9.1.3 Results of Findings

- ❖ **Availability of Medical Laboratory Services:** Upon investigation one of the major things we observed was the presence of only two laboratories in the entire community which included a private laboratory that was totally non- functional and the laboratory in the Primary Health Care center (PHC) which lacked histopathology tests totally and other essential tests in chemical pathology, hematology and microbiology. This greatly reduced the community's access to these particular missing services.
- ❖ **Awareness Level :** Upon the analysis of the gathered data, it was discovered that while a considerable percentage of the population demonstrated awareness of medical laboratory services, a notable fraction remained uninformed. Even within the aware group, a subset reported not utilizing these services probably due to issues such as long turnaround time as mentioned by some, further limiting access to essential diagnostics.

9.1.4 Recommendations

- ❖ More medical laboratories should be built, equipped and provided with skilled medical laboratory scientists.
- ❖ The private medical laboratory should be renovated and made functional.
- ❖ The medical laboratory in the primary health care center should be equipped with the missing services.
- ❖ More advocacy for medical laboratory services should be carried out in the community to improve the level of awareness of these services.

9.1.5 Conclusion

The level of access to the medical laboratory services in the community is below average especially due to the presence of only one functional laboratory in the entire community. However, the importance of laboratory services in complete healthcare can not

be overemphasized and by implementing the recommendations, a significant change would be observed in the access of the people to medical laboratory services.

9.2 MLS MLO 2

TO IDENTIFY THE GAPS IN THE PROVISION OF EFFECTIVE MEDICAL LABORATORY SERVICES IN THE COMMUNITY

9.2.1 Introduction

Medical laboratory services encompasses a wide range of diagnostic and analytical techniques which are procedures that play a crucial role in patient care, early disease diagnosis, and public health promotion. However, there are gaps or challenges which hinder the provision of effective medical laboratory services. These issues not only hinder the efficiency of laboratory services but also have far-reaching consequences for healthcare delivery. It is therefore important to identify the gaps in the provision of medical laboratory services in Shao community.

9.2.2 Methodology

We visited the Primary Health Care Center, the only facility in the area with a functional medical laboratory. The visit involved direct observation of laboratory operations and interactive discussions with the laboratory staff. This approach provided firsthand insights into the available diagnostic services, operational challenges, and the role of medical laboratory scientists in patient care.

9.2.3 Results of Findings

❖ **Lack Of Infrastructure:** Electricity being a major infrastructure needed for the functioning of the medical laboratory especially in terms of powering equipment such as centrifuge, is a major challenge in Shao. The electricity supply is bad as once a month based on our interaction with the staff at the health facility. Despite this there is

no generator to substitute in the provision of electricity. Although there is solar power, this is not a guarantee that it can power all equal equals especially if more are added.

- ❖ **Workforce Shortage:** We also discovered that in the entire community of Shao, there is only one medical laboratory scientist who works in the primary health care center. This workforce shortage in medical laboratory services creates a significant gap by affecting the efficiency, accuracy, and overall quality of healthcare delivery. Such shortage leads to delay in diagnosis and treatment, increased workload and burnout, hindered research and innovation as this staff would be too occupied with the tests requested.
- ❖ **Financial Constraints/Inadequate Funding:** The lack of funds invested into the medical laboratory in the community has led to the lack of purchasing power for reagents and other necessary laboratory equipment. On our visit to the primary health center and the laboratory present there, we discovered that despite having microscopes provided by a Non- Governmental Organization (NGO), the reagents like giemsa stain, leishmann stain and others were totally absent which has created a huge gap in diagnosis, monitoring and treatment especially in the diagnosis of malaria which is endemic in the community as in many places in Nigeria as well. Even the only available rapid diagnostic kit for malaria was specific to *Plasmodium falciparum*, leaving infections caused by other malaria species undiagnosed. This highlights a critical gap in medical laboratory services—limited diagnostic capacity, which can lead to misdiagnosis, delayed treatment, and inadequate disease surveillance. Without comprehensive testing options, non-falciparum malaria cases may go undetected, contributing to continued transmission and poor patient outcomes.
- ❖ **Limited Government Intervention:** The primary health care center, which is government-owned, includes the only functional laboratory in Shao that, by extension,

is also government-owned has received limited government intervention in terms of funding and equipment supply. Ordinarily, the provision of essential laboratory equipment should be the responsibility of the government, but this has not been the case. Instead, much of the laboratory's infrastructure has been sustained through donations from non-governmental organizations, such as the HYGIEA Foundation, which provided microscopes and a solar panel. This reliance on external support underscores the gaps in government involvement in strengthening laboratory services in the community.

9.2.4 Recommendations

- ❖ Alternative sources of power supply should be provided such as a generator or solar panel that is able to power all necessary equipment in the laboratory.
- ❖ Employment of more medical laboratory scientists in the laboratory which will serve the community.
- ❖ Sourcing and allocation of sufficient funds by the government and private organizations to the areas of the medical laboratory services that are lacking.
- ❖ The government should be more involved in the enhancement of medical laboratory services especially in the primary health care center through the provision of equipment, funding, infrastructure etc which will in turn benefit the populace.

9.2.5 Conclusion

There are major gaps in the provision of effective laboratory services in the community which are glaring, however these gaps can be effectively filled if the proper measures and recommendations are implemented. These measures will eventually strengthen medical laboratory services in Shao community, improve healthcare delivery, reduce disease burdens, and ensure a healthier population.

9.3 MLS MLO 3

TO IDENTIFY BARRIERS TO ACCESSING MEDICAL LABORATORY SERVICES IN THE SHAO COMMUNITY.

9.3.1 Introduction

Access to medical laboratory services is a crucial component of effective healthcare delivery, playing a significant role in disease diagnosis, treatment monitoring, and public health interventions (WHO, 2021). However, in many under-served communities, including SHAO, several factors hinder access to these essential services. A community evaluation revealed that lack of awareness, unavailability of equipment and reagents for specific tests, inadequate facilities, and shortage of personnel are the primary barriers. Addressing these issues is vital to improving healthcare outcomes and ensuring that the population benefits from timely and accurate laboratory diagnostics.

9.3.2 Results of Findings

- ❖ **Lack of Awareness:** One of the significant challenges affecting access to medical laboratory services in SHAO is the lack of awareness among residents. Many community members are unaware of the importance of laboratory tests in diagnosing and managing diseases. This low level of health literacy often results in delayed medical consultations and reliance on self-medication or traditional remedies, which may not be effective . Increasing public awareness through health education campaigns and community outreach programs can encourage individuals to utilize laboratory services and seek timely medical care.
- ❖ **Cultural Treatment Preferences:** A significant portion of the Shao community prefers traditional forms of medication passed down through generations. They believe some of the measures employed by their predecessor worked and kept them healthy, so they tend to respect such practices and believe going against it is a form of taboo.

Despite advancements in modern diagnostic medicine, these cultural beliefs persist, leading residents to rely on unconventional treatments relying solely on signs and symptoms of diseases without any form of testing or diagnosis.

- ❖ **Lack of Trust & Confidence in Diagnostic Services:** As a result of bad practices within the available laboratories, previous experiences or misinformation spread within the communities, members of the communities have lost trust and/or confidences in the laboratory services offered. They either resort to self diagnosis and medication or journey outside the town for daignosis.
- ❖ **Inadequate number of laboratories:** During our survey of the communities, we discovered that only 2 laboratories exists within shao community with over 60,000 population. Factors such as distance to laboratory and over crowding have discouraged individuals from assessing local diagnostic seervices.

9.3.3 Conclusion

The inaccessibility of medical laboratory services in the SHAO community is a multifaceted issue that requires urgent attention. Factors such as lack of awareness, distance, lack of trust and confidence in diagnosis and insufficient personnel all contribute to the problem. Addressing these barriers through public health education, improved healthcare investment, and workforce development will enhance laboratory accessibility and improve overall healthcare outcomes. Policymakers and stakeholders must work together to implement sustainable solutions to ensure equitable access to medical laboratory services in SHAO.

OPTIONAL PROJECT

COMMUNITY SERVICE: SHAO SANITATION DRIVE

10.1 Introduction

On the 22nd of march, 2025, the entire class headed out to clean the community that was polluted with plastics, nylon bags and other refuse.



Figure 10.1: Sanitation of Shao community

10.2 Selected Areas for Cleaning

- ❖ The street leading to the Unilorin COBES Hostel.
- ❖ Streeting leading to the baptist church
- ❖ The market area.
- ❖ Gutters by the King's Palace.
- ❖ Gutter in front of the Ajakitipa of Shao compound.

10.3 Reasons for Selecting those Sites

- ❖ To prevent the breakout of food-borne illnesses.
- ❖ To prevent the breeding areas for mosquitoes causing malaria.
- ❖ To improve the living conditions of people within these areas

10.4 Summary of Procedure

The sanitation started directly in front of the COBES site and ended in 2 locations, in front of the King's Palace and in front of the baptist church. We divided the group into 2 and went in opposite directions at the junction leading to the COBES site cleaning and sanitizing these areas as we went.

10.5 Challenges

- ❖ Lack of proper equipment
- ❖ The community was heavily littered
- ❖ No proper site for incineration of collected waste
- ❖ Distance from the sanitation sites to the dumping sites were too far apart

10.6 Conclusion

It was noted that the locals in the community of Shao are not very conscious of the cleanliness of their environment. We advise that the locals take serious note of the sanitary conditions of their environment as it can help in curbing various diseases at their infancy stage and prevent them from spreading.

OPTIONAL STUDIES

KNOWLEDGE AND SKILLS ACQUISITION

The town of Shao is celebrated for its mastery in producing staple foods such as dried okro powder (locally known as orunla), garri, and beske. As part of our community engagement initiatives, we took the opportunity to immerse ourselves in the traditional production processes of these foods, thereby deepening our understanding and appreciation of the local culture.

11.1 Garri Making

Garri (cassava flakes) is one of the many food products that can be made from fresh cassava roots. It is a dry, crispy, creamy-white, and granular product which is widely consumed in West Africa (James *et al.*, 2016).

On the 27th of March, 2025, we proceeded to learn some of the local foods made in Shao, one of which is “garri”. We went to Shao cassava market, Isale Oja where we saw the different stalls in which garri processing took place. On getting there we were quickly greeted with the sight of hardworking women who were frying garri, after this we met the owner who explained the process of garri making and showed us each process taking place in the different stalls and we then proceeded to make garri ourselves.

Some of us peeled the fresh cassava roots, while others blended the already peeled cassava roots which was then poured into large sacks for the water and then pressed to remove excess water. The pressed garri is then left to ferment for three to five days depending on the weather, in dry/hot seasons it can be left for three days but is left for five days in other seasons. This step is crucial to get rid of cyanide present in the root of cassava. After this, the garri is filtered, fried thoroughly and then sun dried once again. After this, it is packaged for sale or consumption.

11.1.1 Steps for Garri Production/ Processing

- ❖ Harvest fresh cassava roots
- ❖ Peel the cassava roots
- ❖ Grind the cassava roots into tiny shafts using a specialized grinding machine.
- ❖ Package the grinded cassava into big sacks (rice sacks, sugar sacks).
- ❖ Press out the water from the cassava using a jack.
- ❖ Allow it to ferment for three to five days to remove cyanide.
- ❖ Filter it using a machine or manual method.
- ❖ Fry the garri thoroughly so that every grit is separated.
- ❖ Sundry the fried garri.
- ❖ Package the garri properly.

We also learned that the shaft filtered out from the garri can also be grinded further to a very smooth consistency to get a powdery form called 'lebu'. Also another form of garri can be made by the addition of palm oil to the garri while grinding it initially which will give an even colour or it can be added while frying, however this method may not produce an even colour and it consumes a lot of palm oil.

Garri production in Shao showcases the entrepreneurial skills of the people. Although the process can be demanding, the satisfaction of achieving the final product makes it worthwhile.

11.2 Beske Production

Beske (Soyabean curd) is a popular Nigerian dish that translates to “fried tofu” or “fried soyabean curd”. Beske, a tofu-like cheese made from soyabean milk, is a good source of protein, fats, and minerals like calcium, iron, and phosphorus, making it a valuable food in the diet.

On the 27th of March, 2025, we proceeded to learn some of the local foods made in Shao, one of which is “Beske.” We went to a Beske maker at Isale Oja where we were enlightened on the process of Beske making. She was generous enough to show us the materials and equipments used in the process along with already made Beske that was currently been packed for sale. The process was carefully explained to us alongside tips and tricks on getting the best product. We asked questions after her explanation and she also asked us questions to ensure we understood correctly.

11.2.1 Steps for Beske Production/ Processing

- ❖ Soyabeans are purchased and picked to ensure it is free of impurities
- ❖ Picked Soyabeans are soaked about 1 hour before preparation
- ❖ After 45 minutes of soaking, soaked Soyabeans are blended
- ❖ The grounded Soyabean paste is boiled slowly while adding water and slightly fermented Soya milk called “Omi Dun,” to facilitate curding. The fermented Soya milk serves as a coagulant.
- ❖ A cloth sieve is used to separate the curd from the boiling mixture
- ❖ The curd is pressed by placing the sieve cloth containing it on a flat surface (wood/rock) and putting a heavy substance on it.
- ❖ The pressed curd results into Beske also known as. Tofu or Soyabean curd
- ❖ The curd is cut into desired shape.
- ❖ Beske can be fried, roasted or marinated in peppered sauce for more flavor.



Figure 11.1: Skill acquisition; Garri and Beske preparation.

PROBLEMS, SOLUTIONS AND RECOMMENDATIONS

12.1 Problems Encountered

The following are the problems encountered in the course of carrying out our Minimum Learning Objectives (MLOS)

- ❖ **Illiteracy:** A significant portion of the community members were elderly and uneducated, it was difficult to get information through from them especially during our socio-demographic study.
- ❖ **Language barrier:** Shao is a multicultural and multilingual community where various languages like Yoruba (major language), Hausa, Tapa/Nupe and Igbo is spoken. As a result, we could not interact with all the members of the community.
- ❖ **Receptiveness:** Some community members were not open to discussion, as they felt like it was a waste of time. Some community members were hesitant to answer some certain questions. They had reservations on what questions to answer and what not to, and they believed some of the questions were personal.
- ❖ **Network issues:** The questionnaire used for data collection and community survey was an online based form. The Google form was created to ease data access and transfer and to make the collection easier. However, certain places in the community had no network reception which impeded our progress.
- ❖ **Time constrain:** The duration of the was quite short, leading to a rush in decision making and lack of proper deliberation of key issues in achieving the minimum learning objectives.

12.2 Solutions Proffered

- ❖ In respect to the language barrier, members of the group who could speak some of the major languages were deployed to speak. It aided our interactions with them and made data collection easier.

- ❖ We respectfully introduced ourselves and explained the reason we came to their community. Our ID cards and lab coats convinced them. This made the members of the community more receptive.
- ❖ We helped the community members in filling the questionnaire, this aided information collection.
- ❖ We divided ourselves into different groups which worked simultaneously on different aspects of the MLO to make up for the lack of time.

12.3 Recommendations

Based on our experiences and findings from the Community-Based Experience and Services (COBES) program in Shao community, we propose the following recommendations to enhance healthcare delivery and improve the overall health outcomes in the area:

❖ Improvement of Laboratory Services:

- It is essential to invest in upgrading the existing laboratory facilities in the primary health care center. This can include the procurement of modern diagnostic equipment and ensuring a reliable supply chain for reagents and essential tools.
- Hiring additional qualified medical laboratory scientists will not only improve service delivery but also ensure that the community receives comprehensive diagnostic services.

❖ Capacity Building for Healthcare Workers:

- Continuous training and education programs should be implemented for laboratory staff and healthcare providers to update their skills on standard diagnostic protocols, biosafety measures, and the management of emerging health challenges.
- Workshops that focus on community health education can empower local healthcare workers and promote best practices in patient care.

❖ **Sustainable Water and Sanitation Projects:**

- Initiate and support projects aimed at improving access to clean water by rehabilitating boreholes and ensuring the maintenance of existing water supply systems. This can significantly reduce waterborne diseases prevalent in the community.
- Environmental sanitation should be prioritized by constructing incinerators for proper waste management, thus improving the overall health environment.

❖ **Community Health Education:**

- Develop and implement community health education programs that target prevalent health issues, such as malaria, HIV/AIDS, and maternal health. Educational campaigns should utilize various platforms, including schools, community meetings, and local media.
- Engage community members in health promotion activities to foster a deeper understanding of health practices and encourage behavior change.

❖ **Strengthening Local Health Governance:**

- Encourage collaboration between local government health officials and community leaders to develop health policies that reflect the needs of the population. Regular community health assessments can guide these policies and interventions.
- Facilitate the formation of health committees within the community to ensure active participation of residents in health-related decision-making processes.

❖ **Funding and Resource Allocation:**

- Advocate for increased government and private organization funding to support healthcare initiatives in Shao. Sourcing grants or establishing partnerships with NGOs can provide additional resources for community health projects.

- Prioritize funding for essential health services, including maternal and child healthcare, to ensure vulnerable populations receive adequate attention.

❖ **Future COBES Engagement:**

- Subsequent COBES teams should build on the foundations laid by previous groups. This continuity can ensure that projects initiated will be sustainable and have long-term impacts.
- Future students should engage with the community prior to their arrival to better understand local health challenges and align their projects with the residents' needs.

CONCLUSION

The Community-Based Experience and Services (COBES) program conducted in the Shao community from March 17 to March 28, 2025, has provided invaluable insights into the local healthcare landscape. The engagement of the medical laboratory sciences students with the community highlighted significant gaps in health services and identified critical barriers to accessing quality healthcare. Through the use of structured questionnaires and direct interaction with community members, we were able to assess local health practices and obtained feedback on the perceptions and realities of health service delivery.

The findings underscore the urgent need for enhanced healthcare education and services within the community. Instances of reliance on informal healthcare practitioners revealed a considerable lack of trust and availability in formal health services.

Additionally, our experiential learning has reinforced the significance of community-based interventions in shaping successful health programs. The collaboration between the COBES unit and the Shao community has laid a foundation for ongoing health improvement efforts, empowering residents to take proactive measures towards their health and well-being.

By implementing the proposed recommendations and nurturing ongoing partnerships, it is anticipated that the health outcomes for the Shao community will improve markedly. Overall, the COBES experience serves as a testament to the importance of integrating practical learning with community health engagement, ultimately contributing to a healthier and more informed population.

APPENDIX I

Template Used For Health Care Service Survey

1. What type of healthcare facility is this?
 - PHC
 - Private hospital
 - Pharmacy
 - Clinic
 - Chemist Store
2. What services does this facility provide?
3. How many patients does the facility serve daily?
4. How far is the nearest referral hospital?
5. What transportation options are available for patients to reach this facility?
6. How frequent does the facility have electricity and clean water supply?
7. What are the backup power sources used for electricity and water?
8. Are there adequate sanitation facilities and waste disposal systems for patients and staff? Yes or No
9. What are the most common illnesses or health issues tested & treated here?
10. Are laboratory and diagnostic services available? If yes, what are the services available?
11. What test is most commonly requested?
12. Is there any vaccination program? If so, which vaccines are provided?
13. Are the laboratory equipment and supplies adequate and functional?
14. What challenges do you face in providing quality healthcare services?
15. How many healthcare professionals work here?
16. How many are medical laboratory scientists?

17. Is the current staff sufficient to meet the community's healthcare needs? Yes or No
18. Do staff members receive regular training or skill updates? Yes or No
19. What outreach programs are carried by the facility?
20. What are the governmental or NGO involvement or contribution to this facility?

APPENDIX II

Template Used For Socio Demography Data Collection

1. Household Name/ Serial number
2. Age
3. Sex
4. Occupation
5. Religion
6. Marital Status
7. Ethnicity
8. What is the monthly income in this household?
9. Highest Level of Education
10. Family Size
11. Total number of men, women & children
12. Age range of children in this household
13. Are you currently receiving treatment for any medical condition? (If yes, please specify; If no, "Not applicable")
14. Do you have any recurring or chronic illness(es)? (If yes, please specify; If no, "Not applicable")
15. Were you tested for this illness mentioned above?
16. Health facilities visited/used
 - PHC
 - Missionary Clinic
 - Private Clinic
 - Unlicensed Practitioners
 - Chemist or pharmacy

17. Primary Method of Treatment?
18. What is your reason for choosing these treatment methods? (Select 1 or more)
- Personal belief or cultural traditions
 - Previous positive experience
 - Fewer side effects compared to other treatments
 - Accessibility or affordability
 - Recommendation from family, friends, or community
 - Dissatisfaction with conventional medicine
19. Are you aware of the available health facilities and the services they provide in this area? Yes or No
20. How frequently do you visit the health facilities here?
- Regularly (e.g. monthly or more)
 - Occasionally (a few times a year)
 - Rarely (once a year or less)
 - Only in emergencies
21. How satisfied are you with the healthcare services you have received?
- Very Satisfied
 - Satisfied
 - Neutral
 - Dissatisfied
 - Very dissatisfied
22. Are healthcare initiatives offered by the government/private sector to the citizens of Shao? Yes or No
23. If yes, what are the healthcare initiatives offered by the government/private sector to the citizens of Shao?

- Educational Outreaches
 - Medical/Diagnostic Outreaches
 - Vaccinations
24. What is your source of water supply? (Select 1 or more)
- Borehole
 - Stream
 - Well
25. Are you aware of laboratory services available here in Shao? Yes or No
26. Have you ever used the laboratory services? Yes or No
27. What types of test have you undergone?
28. Were all the tests you needed available at the facility? Yes or No
29. What challenges have faced in accessing laboratory services?
30. What improvements would you suggest for laboratory services in this facility?
31. What type of sewage system is in place?
32. What method of waste disposal is used by the family?
- Municipal garbage trucks
 - Open dumping (unregulated disposal in open areas)
 - Landfills (controlled or uncontrolled)
 - Incineration (burning of waste)
33. Is the government/ private sector involved in the community's waste disposal? Yes or No
34. If yes, how are they involved in the community's waste disposal?
- Government-managed waste collection (e.g., municipal garbage trucks)
 - Private waste collection services
 - Community-based waste management programs

APPENDIX III

Template Used For Result Of Tests

 DEPARTMENT OF MEDICAL LABORATORY SCIENCE, UNIVERSITY OF ILORIN IN COLLABORATION WITH W.H.O.											
PATIENT'S NAME:	AGE	SEX									
HEIGHT:	WEIGHT:	BMI:									
URINALYSIS <table style="width: 100%; border: none;"> <tr> <td style="width: 33%;">LEUCOCYTE _____</td> <td style="width: 33%;">NITRATE _____</td> <td style="width: 33%;">UROBILINOGEN _____</td> </tr> <tr> <td>PROTEIN _____</td> <td>PH _____</td> <td>BLOOD _____</td> </tr> <tr> <td>KETONE _____</td> <td>BILIRUBIN _____</td> <td>GLUCOSE _____</td> </tr> </table>			LEUCOCYTE _____	NITRATE _____	UROBILINOGEN _____	PROTEIN _____	PH _____	BLOOD _____	KETONE _____	BILIRUBIN _____	GLUCOSE _____
LEUCOCYTE _____	NITRATE _____	UROBILINOGEN _____									
PROTEIN _____	PH _____	BLOOD _____									
KETONE _____	BILIRUBIN _____	GLUCOSE _____									
BLOOD PRESSURE: _____ mmHg / _____ mmHg PULSE _____ BPM											
<table style="width: 100%; border: none;"> <tr> <td style="width: 30%;">Malaria RDT</td> <td style="width: 35%;">☐ REACTIVE</td> <td style="width: 35%;">☐ NON-REACTIVE</td> </tr> <tr> <td>HbsAg</td> <td>☐ REACTIVE</td> <td>☐ NON-REACTIVE</td> </tr> <tr> <td>M.Tuberculosis Antibody</td> <td>☐ REACTIVE</td> <td>☐ NON-REACTIVE</td> </tr> </table>		Malaria RDT	☐ REACTIVE	☐ NON-REACTIVE	HbsAg	☐ REACTIVE	☐ NON-REACTIVE	M.Tuberculosis Antibody	☐ REACTIVE	☐ NON-REACTIVE	Comment _____ _____ Scientist's Signature _____
Malaria RDT	☐ REACTIVE	☐ NON-REACTIVE									
HbsAg	☐ REACTIVE	☐ NON-REACTIVE									
M.Tuberculosis Antibody	☐ REACTIVE	☐ NON-REACTIVE									